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This is a corrected copy of the paper cited above. The text is retyped from the printed paper. It keeps the paper's wording and the numbering of its results, equations and footnotes, and a marker such as [p. N] shows where page N of the printed paper begins.

The corrections come from a formalization of the Graph Minors series in the Lean proof assistant, github.com/mahdi-jfri/GraphMinor, in which the results of this paper are proved. Where the formalization found that a statement or a step of a proof is wrong as printed, this copy corrects it in place. Where the paper leaves a proof or a step to the reader, this copy adds it, also in blue. Text ~~struck through in red~~ is removed and text in blue is inserted. Each change has a note, marked F1, F2, ..., that gives the reason. The file corrections/GM09.md of the formalization describes each correction in full. Each note, and each item in the list below, begins with the kind of the change. A **misprint** is a wrong symbol, index or word. A **statement mistake** is a result or a claim of the text that is false as printed. A **degenerate case** is a statement that fails only in a trivial case, such as an empty graph or a parameter equal to 0. A **proof gap** is a step of a proof that does not follow as written. A **missing proof** is a proof that the paper omits, leaves to the reader or only sketches, and that this copy adds.

The corrections and added proofs, in the order of the pages:

1. **Missing proof.** Proof of (3.4), page 48: the proof is added.
2. **Degenerate case.** Statement of (8.1), page 60: the hypothesis that $\bar{\Omega} \neq \emptyset$ or G is null is added.
3. **Missing proof.** Proof of (10.1), page 64: the omitted proof is added.
4. **Proof gap.** Proof of (10.2), page 64: Q_i runs along C_{s+1-i} in place of C_i, and the arc of the circuit is chosen by the side on which its disc lies.
5. **Proof gap.** Proof of (10.3), page 64: in place of passing to $\Omega \mid (\bar{\Omega} - \{x\})$, the induction keeps the neighborhood and shrinks a set Y that holds the ends.
6. **Proof gap.** Proof of (10.3), page 65: (5.2) is used in a form that allows the paths Q_i to have internal vertices in $\bar{\Omega}$, and this form is derived from (5.2).

7. **Misprint.** Proof of (10.4), page 66: "P^+ is a path of G" becomes "P^+ is a path of $G \cup G_0$".
8. **Missing proof.** Proof of (10.4), page 66: the reason why the extended paths form a crooked transaction is added.
9. **Missing proof.** Proof of (11.1), page 67: the omitted proof is added.

[p. 40] Graph Minors. IX. Disjoint Crossed Paths

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Let G be a graph, with a cyclic order imposed on a subset of its vertices (called "special" vertices). We show that either
(i) modulo (≤ 3)-separations, G can be drawn in a disc with no crossings except in one "small" area, and with its special vertices on the outside in the correct order, or
(ii) there is a partition of the special vertices into two "semicircles," and there is a large collection of vertex-disjoint paths of G running from one semicircle to the other, such that each of these paths is either crossed by another or lies between two others.
This is basically a lemma to be used in later papers.

1. Introduction

We hope to prove in a later paper that for any graph H, every graph with no minor isomorphic to H can be constructed by piecing together in a tree structure graphs which can "almost" be drawn in surfaces of bounded genus. This result appears to have a number of important consequences, including the truth of Wagner's well-quasi-ordering conjecture for graph minors, and the existence of a polynomial algorithm for the Disjoint Connecting Paths problem. However, the proof of this embedding structure theorem is long, and it is convenient to present it in sections. Here we [p. 41] develop some of the theory behind the use of "almost" in the statement of the theorem.

Let Ω be a cyclic permutation of the elements of some finite set; we denote this set by $\bar{\Omega}$. We denote the image under Ω of $x \in \bar{\Omega}$ by $\Omega(x)$. If $X \subseteq \bar{\Omega}$, Ω induces a cyclic permutation Ω' of X in the natural way—that is, for $x \in X$ we define $\Omega'(x)$ to be the first term of the sequence $\Omega(x)$, $\Omega(\Omega(x))$, ... which is in X. We define $\Omega \mid X = \Omega'$.

Let $v_1, ..., v_k \in \bar{\Omega}$ be distinct. We say that $(v_1, v_2, ..., v_k)$ is Ω-*clockwise* (or more briefly, *clockwise*, when the dependence on Ω is to be understood) if $\Omega'(v_1) = v_2$, $\Omega'(v_2) = v_3$, ..., $\Omega'(v_k) = v_1$ where $\Omega' = \Omega \mid X$. If $v_1, v_2, v_3, v_4 \in$

*This research was performed under a consulting agreement with Bellcore.

$\bar{\Omega}$ are distinct and one of (v_1, v_2, v_3, v_4), (v_4, v_3, v_2, v_1) is clockwise, we say that $\{v_1, v_3\}$ *crosses* $\{v_2, v_4\}$. If $v_1, v_2 \in \bar{\Omega}$ are distinct, we define

$$\Omega(v_1, v_2) = \{v \in \bar{\Omega} - \{v_1, v_2\} : (v_1, v, v_2) \text{ is clockwise}\}.$$

Then $(\Omega(v_1, v_2), \Omega(v_2, v_1), \{v_1, v_2\})$ is a partition of $\bar{\Omega}$. A fence of Ω is a pair (A, B) of subsets of $\bar{\Omega}$ such that $A \cap B = \emptyset$, $A \cup B = \bar{\Omega}$, and for distinct $a_1, a_2 \in A$ and distinct $b_1, b_2 \in B$, $\{a_1, a_2\}$ does not cross $\{b_1, b_2\}$. Thus, intuitively, a fence is a partition of $\bar{\Omega}$ into two "segments." If V is a set, a cyclic permutation Ω is said to be *in* V if $\bar{\Omega} \subseteq V$.

Graphs in this paper are finite, and may have loops or multiple edges. If G_1 is a subgraph of G we write $G_1 \subseteq G$. If $G_1, G_2 \subseteq G$, we define the subgraphs $G_1 \cup G_2$, $G_1 \cap G_2$ by

$$G_1 \cup G_2 = (V(G_1) \cup V(G_2), E(G_1) \cup E(G_2))$$
$$G_1 \cap G_2 = (V(G_1) \cap V(G_2), E(G_1) \cap E(G_2)).$$

A society is a pair (G, Ω) where G is a graph and Ω is a cyclic permutation in $V(G)$. If (G, Ω) is a society we say that P is an Ω-*path* if P is a path in G with ends distinct and both in $\bar{\Omega}$. If $\mathcal{P}$ is a set of mutually vertex-disjoint Ω-paths, we say that a fence (A, B) of Ω is a *fence for* $\mathcal{P}$ if each $P \in \mathcal{P}$ has one end in A and the other in B; and if there is a fence for $\mathcal{P}$ we say that $\mathcal{P}$ is a *transaction in* (G, Ω).

Let $\mathcal{P}$ be a transaction in (G, Ω). We denote by $V(\mathcal{P})$ the set of all ends of all members of $\mathcal{P}$, and by $\bigcup \mathcal{P}$ the subgraph of G which is the union of all members of $\mathcal{P}$. Let $P \in \mathcal{P}$ have ends a, b. We say that P is *peripheral in* $\mathcal{P}$ if one of $\Omega(a, b) \cap V(\mathcal{P})$, $\Omega(b, a) \cap V(\mathcal{P})$ is empty. It is easy to see that any transaction has at most two peripheral members. A transaction with no peripheral members is said to be *crooked*.

The object of this paper is to prove a theorem concerning the structure of societies in which there is no large crooked transaction. If a society can be drawn without crossings in a disc with the vertices of $\bar{\Omega}$ on the outside in the correct order, it can have no non-null crooked transaction. (This [p. 42] result has a converse, which is a prototype of our main theorem, and which we discuss in Section 2.) Similarly, a society which can be drawn in the above way except for one crossing may have a crooked transaction of cardinality two, but has none of cardinality three or more. In contrast, a society with two crossings, sufficiently far apart, may admit a large crooked transaction. But in general, if a society can be drawn in a disc so that all its crossings are pairwise close together (in a sense which we shall specify in a moment) then it has no large crooked transaction. The main theorem of

this paper is a converse; that such a society is (modulo (≤ 3)-separations) the most general without a large crooked transaction.

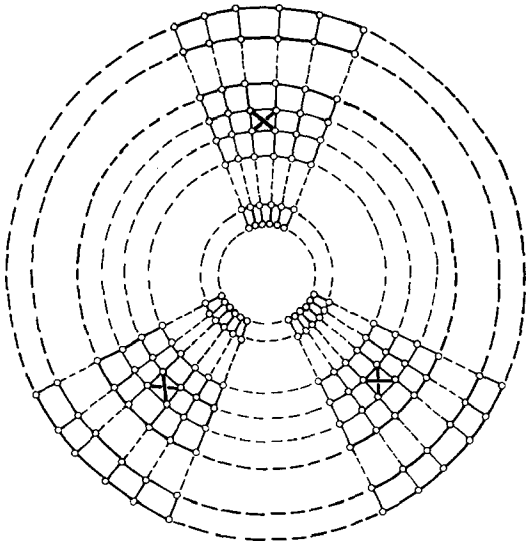
Let us make the above ideas precise. We say that (G, Ω, Ω_0) is a *neighborhood* if G is a graph and Ω, Ω_0 are both cyclic permutations in $V(G)$. Let Σ be a plane, with some orientation called "clockwise." We say that a neighborhood (G, Ω, Ω_0) is *rural* if G has a drawing Γ in Σ without crossings (so G is planar) and there are closed discs $\Delta_0 \subseteq \Delta \subseteq \Sigma$, such that

(i) the drawing Γ uses no point of Σ outside Δ, and none in the interior of Δ_0, and

(ii) for $v \in V(G)$, the point of Σ representing v in the drawing Γ lies in $bd(\Delta)$ (respectively, $bd(\Delta_0)$) if and only if $v \in \bar{\Omega}$ (respectively, $v \in \bar{\Omega}_0$), and the cyclic permutation of $\bar{\Omega}$ (respectively, $\bar{\Omega}_0$) obtained from the clockwise orientation of $bd(\Delta)$ (respectively, $bd(\Delta_0)$) coincides (in the natural sense) with Ω (respectively, Ω_0).

A separation of a graph G is a pair of subgraphs (G_1, G_2), where $G_1 \cup G_2 = G$ and $E(G_1 \cap G_2) = \emptyset$. A separation of a society (G, Ω) is a separation (G_1, G_2) of G such that $\bar{\Omega} \subseteq V(G_2)$. Let (G_1, G_2) be a separation of (G, Ω), and let Ω_1 be a cyclic permutation with $\bar{\Omega}_1 = V(G_1 \cap G_2)$. Then we say that (G, Ω) is the *composition* of the society (G_1, Ω_1) with the neighborhood (G_2, Ω, Ω_1). The *sophistication* of (G, Ω) is the minimum $p \geq 0$ such that (G, Ω) is the composition of a society which has no transaction of cardinality $> p$ with a rural neighborhood. It is easy to prove (directly, or from (9.1)) that if (G, Ω) has sophistication $\leq p$ then it has no crooked transaction of cardinality $> p$. The converse, however, is false. Let G be the graph obtained from a cylinder graph with $3p$ radial lines and a large number of nested circles by adding three crossing pairs of edges as in Fig. 1. Let Ω be the cyclic permutation of the vertices of G on the outside circle. It can be shown that (G, Ω) has no crooked transaction of cardinality $> p + 2$, and yet its sophistication is at least $\lfloor \frac{3}{2} p \rfloor$.

The main result of this paper is nevertheless a partial converse. We show that modulo (≤ 3)-separations, a society in which there is no crooked transaction of cardinality $\geq p (\geq 4)$ has sophistication $\leq 3p + 9$.



[p. 43]

FIGURE 1

2. Two Crossing Paths

The case when (G, Ω) has no non-null crooked transaction is of special importance, and in essence has previously been characterized [3-6]. However, the methods of this paper give a new proof of the characterization which is simpler than those already published, and in this section we give this proof. This is not much of a digression from the proof of the main theorem, because we need the lemmas that we give here for the main proof in any case.

If P is a path in a graph G and $Z \subseteq V(G)$ with $Z \cap V(P) = \emptyset$, we say that P *avoids* Z. We denote by $G \setminus F$ the graph obtained from G by deleting F. (Here F may be a vertex or a set of vertices, or an edge or a set of edges.) We say that Z *catches* G if $Z \subseteq V(G)$ and $Z \cap V(G') \neq \emptyset$ for every component G' of G.

(2.1) *Let G be a simple graph and let $Z \subseteq V(G)$. Let $s, t \in V(G)$ be distinct and such that there is a path between s and t avoiding Z. Choose a subgraph H of G maximal such that Z catches H, and there is a path of G between s and t avoiding $V(H)$. Let $K = G \setminus V(H)$, and let P be a path of K between s and t. Then either $P = K$, or there is a separation (G_1, G_2) of G with $\{s, t\} \subseteq V(G_1)$, $|V(G_1 \cap G_2)| = 2$, $E(P \cap G_2) \neq \emptyset$ (and hence $G_1 \cap G_2 \subseteq P$), $H \subseteq G_1$, and $V(G_1) \neq V(G)$.*

[p. 44] *Proof.* Let $X \subseteq V(K)$ be the set of all vertices of K which are adjacent in G to some vertex of H. We observe, from the maximality of H,

that

(1) *Every path of K between s and t passes through every vertex of X.*

In particular $X \subseteq V(P)$; let $X \cup \{s,t\} = \{x_0, x_1, \ldots, x_k\}$, enumerated in order of occurrence in P, where $s = x_0$ and $t = x_k$. From (1), there are subgraphs $K_1, \ldots, K_k$ of K with union K, such that for $1 \le i \le k$ $K_i \cap P$ is the subpath of P between x_{i-1} and x_i, and for $1 \le i \le k-1$ $V(K_i \cap K_{i+1}) = \{x_i\}$, and for $1 \le i, j \le k$ with $j \ge i+2$, $V(K_i \cap K_j) = \emptyset$. If $V(K_i) = \{x_{i-1}, x_i\}$ for all i $(1 \le i \le k)$ then $P = K$, and so we may assume that $V(K_i) \ne \{x_{i-1}, x_i\}$ for some i. Let $G_2 = K_i$ and $G_1 = G \setminus (V(K_i) - \{x_{i-1}, x_i\}) \setminus E(K_i)$. Then (G_1, G_2) is a separation satisfying the theorem. □

Two paths of a graph G are *disjoint* if they have no common vertex. Let G be a graph and let $X, Y \subseteq V(G)$. By an $X \to Y$ path in G we mean a path with initial vertex and no other in X, and with terminal vertex and no other in Y. Let $X' \subseteq X$ and $Y' \subseteq Y$ be such that $|X'| = |Y'|$. We write $X' \Rightarrow Y'$ (with respect to X, Y) if there are $|X'|$ mutually disjoint $X \to Y$ paths of G from X' to Y'. Our second lemma is a standard result in network flow theory, and can easily be proved by an "augmenting path" argument.

(2.2) *Let G be a graph and let $X, Y \subseteq V(G)$. Let $k \ge 0$ be an integer such that there are k mutually disjoint $X \to Y$ paths. Let $X' \subseteq X$, $Y' \subseteq Y$ be such that $|X'| = |Y'| \le k$ and such that $X' \Rightarrow Y'$ with respect to X, Y. Then there exist $X'' \subseteq X$ and $Y'' \subseteq Y$ such that $X' \subseteq X''$, $Y' \subseteq Y''$, $|X''| = |Y''| = k$ and $X'' \Rightarrow Y''$ with respect to X, Y.*

Let (G, Ω) be a society, and let (G_1, G_2) be a separation of (G, Ω). Let $k \ge 0$ be an integer. We say that (G_1, G_2) is a *k-separation* of (G, Ω) if $|V(G_1 \cap G_2)| \le k$, and $V(G_2) \ne V(G)$. A k-separation is *internal* if $|V(G)| \ge |V(G_2)| + 2$. We say that (G, Ω) is *k-connected* if it has no k'-separation for $k' < k$, and *internally k-connected* if it has no internal k'-separation for $k' < k$.

Let P_1, P_2 be disjoint Ω-paths in (G, Ω), where P_i has ends s_i, t_i $(i = 1, 2)$. If $\{s_1, t_1\}$ crosses $\{s_2, t_2\}$, we say that P_1 *crosses* P_2, and that $\{P_1, P_2\}$ is a *cross*. Now let u, v be distinct vertices of G, and let P_1, P_2, P_3 all be paths between u and v, mutually disjoint except for u and v. We say that $\{P_1, P_2, P_3\}$ is a *tripod* in (G, Ω) if there is an internal vertex s_i of P_i $(i = 1, 2, 3)$, and there exist $t_1, t_2, t_3 \in \bar{\Omega}$ such that $\{s_1, s_2, s_3\} \Rightarrow \{t_1, t_2, t_3\}$ with respect to $V(P_1 \cup P_2 \cup P_3), \bar{\Omega}$.

(2.3) *Suppose that $\{P_1, P_2, P_3\}$ is a tripod in (G, Ω). Then either (G, Ω)* [p. 45] *has a cross, or there is an internal 3-separation (G_1, G_2) of (G, Ω) with $P_1 \cup P_2 \cup P_3 \subseteq G_1$.**

Proof. Let $s_1, s_2, s_3, t_1, t_2, t_3$ be as in the definition of a tripod. If there exist four mutually disjoint paths of G between $V(P_1 \cup P_2 \cup P_3)$ and $\bar{\Omega}$, then by (2.2) we may choose them to be $V(P_1 \cup P_2 \cup P_3) \to \bar{\Omega}$ paths, and to have s_1, s_2, s_3 among their initial vertices. But then (G, Ω) has a cross as is easily seen. If there do not exist four such paths, then by Menger's theorem (G, Ω) has a 3-separation (G_1, G_2) with $P_1 \cup P_2 \cup P_3 \subseteq G_1$; and this 3-separation is internal, since $|V(P_1 \cup P_2 \cup P_3)| \geq 5$. □

A society (G, Ω) is rural if (G, Ω, Ω_0) is rural where $\bar{\Omega}_0 = \emptyset$. The characterization of [3-6] essentially asserts that every internally 4-connected society with no cross is rural. Since this may easily be derived from the 3-connected, internally 4-connected case, the characterization is implied by (2.3) and the following.

(2.4) *Let (G, Ω) be a 3-connected society with no cross or tripod. Then (G, Ω) is rural.*

Proof. We proceed by induction on $|E(G)|$, and may therefore assume that G is simple and connected. We assume that $|\bar{\Omega}| \geq 3$, for otherwise the result is trivial. We may choose distinct $s, t \in \bar{\Omega}$ such that there is a path between s and t avoiding Z, where $Z = \bar{\Omega} - \{s, t\}$. We apply (2.1), and deduce that there is a path P joining s and t such that P is an induced subgraph of G and Z catches $G \setminus V(P)$. In particular, no internal vertex of P is in $\bar{\Omega}$.

Put $W_1 = \Omega(s, t)$ and $W_2 = \Omega(t, s)$. For $i = 1, 2$, let H_i be the union of those components of $G \setminus V(P)$ of which some vertex is in W_i. Then $H_1 \cup H_2 = G \setminus V(P)$ since Z catches $G \setminus V(P)$. Moreover, $H_1 \cap H_2$ is null, since (G, Ω) has no cross.

Let Ω_1 be the cyclic permutation in $V(G)$ such that $\bar{\Omega}_1 = W_1 \cup V(P)$, such that

$$\Omega_1 \mid (W_1 \cup \{s, t\}) = \Omega \mid (W_1 \cup \{s, t\})$$

and such that the sequence of vertices obtained by following P from t to s is Ω_1-clockwise. Define Ω_2 similarly, where $\bar{\Omega}_2 = W_2 \cup V(P)$, and the sequence of vertices obtained by following P from s to t is Ω_2-clockwise. Let $G_1 = (G \setminus E(P)) \setminus V(H_2)$, $G_2 = (G \setminus E(P)) \setminus V(H_1)$. Then (G_1, Ω_1), (G_2, Ω_2) are 3-connected societies, both with fewer edges than (G, Ω).

Suppose that $\{P_1, P_2, P_3\}$ is a tripod in (G_1, Ω_1). For $i = 1, 2, 3$, let Q_i be a $V(P_1 \cup P_2 \cup P_3) \to \bar{\Omega}_1$ path of G_1 with ends s_i, t_i where s_i is an internal vertex of P_i and $t_i \in \bar{\Omega}_1$, chosen so that Q_1, Q_2, Q_3 are mutually disjoint.

[p. 46] Since (G, Ω) has no tripod, $t_1, t_2, t_3 \in V(P)$. Since P is induced it follows that

$$V(P_1 \cup P_2 \cup P_3 \cup Q_1 \cup Q_2 \cup Q_3) \cap V(H_1) \neq \emptyset,$$

and so there is a $V(P_1 \cup P_2 \cup P_3 \cup Q_1 \cup Q_2 \cup Q_3) \to W_1$ path Q in G_1 which avoids $V(P)$. For $i = 1, 2$, or 3 there is no end of Q in $V(P_i \cup Q_i)$ which is not an end of P_i, for $\{P_1, P_2, P_3\}$ is not a tripod in (G, Ω). Thus Q, P_1, P_2, P_3 have a common end; but again it follows easily that there is a tripod in (G, Ω), a contradiction. Thus (G_1, Ω_1) has no tripod.

We claim also that (G_1, Ω_1) has no cross. For suppose that $\{P_1, P_2\}$ is a cross of (G_1, Ω_1), where P_i has ends s_i, t_i $(i = 1, 2)$; and let us choose $\{P_1, P_2\}$ so that r is maximum, where $|\{s_1, t_1, s_2, t_2\} - V(P)| = r$. If $r \geq 2$ then (G, Ω) has a cross, as is easily seen, a contradiction. We suppose that $r = 1$, and without loss of generality that $s_1, s_2, t_1 \in V(P)$ and $t_2 \notin V(P)$. Now $V(P_1) \not\subseteq V(P)$ since P is an induced subgraph of G, and so there is a $V(P_1) \to W_1$ path P_3 in G_1 avoiding $V(P)$. This does not avoid P_2, since (G, Ω) has no cross. But then (G, Ω) has a tripod, a contradiction. We deduce that $r = 0$, so that $s_1, s_2, t_1, t_2 \in V(P)$. But there is a path of G_1 between $V(P_1 \cup P_2)$ and W_1 avoiding $V(P)$; and this contradicts the maximality of r.

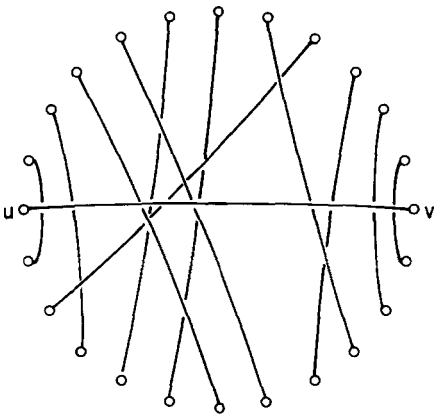
Thus (G_1, Ω_1) has no cross or tripod. But it has fewer edges than (G, Ω), and so it is rural, by our inductive hypothesis. Similarly (G_2, Ω_2) is rural, and it follows that (G, Ω) is rural, as required. □

3. Attachments

We now return to the general problem. Crooked transactions are quite easy to handle because of a certain "exchange" property, which we study in this section.

Let $\mathscr{P}$ be a transaction in (G, Ω), and let P be a $V(\bigcup \mathscr{P}) \to (\bar{\Omega} - V(\mathscr{P}))$ path in G. We call P an *attachment* to $\mathscr{P}$. The common vertex of $\bigcup \mathscr{P}$ and P is called the *foot* of the attachment, while the member P_1 of $\mathscr{P}$ of which the foot is a vertex is called the *leg*. The union of P with a subpath of P_1 from the foot to an end of P_1 is called a *switch*. If $\mathscr{P}$ is crooked, the attachment P is *exchangeable* if there is a switch S such that $(\mathscr{P} - \{P_1\}) \cup \{S\}$ is a crooked transaction. It turns out that attachments are normally exchangeable; one of the results of this section is basically a characterization of non-exchangeable attachments.

Let $\mathscr{P}$ be a crooked transaction and let $P \in \mathscr{P}$. If $\mathscr{P} - \{P\}$ has two peripheral members we say that P is an *overpass* of $\mathscr{P}$. Figure 2 shows a typical overpass.



[p. 47]

FIG. 2. The u-v path is an overpass.

(3.1) *Any crooked transaction has at most one overpass.*

Proof. Let $\mathscr{P}$ be a crooked transaction, let $P_1 \in \mathscr{P}$ be an overpass, and let $P_2 \in \mathscr{P} - \{P_1\}$. Let P_3 be a peripheral member of $\mathscr{P} - \{P_1\}$ different from P_2. Then P_2 does not cross P_3; but every overpass crosses every other member of $\mathscr{P}$, and so P_2 is not an overpass. The result follows. □

If P is an attachment to $\mathscr{P}$ with leg an overpass of $\mathscr{P}$, we call it an *overpass attachment.*

(3.2) *Every attachment to a crooked transaction that is not exchangeable is an overpass attachment.*

Proof. Let P be an attachment to $\mathscr{P}$ in (G, Ω) which is not exchangeable, where $\mathscr{P}$ is crooked, and let (A, B) be a fence for $\mathscr{P}$. Let v be the foot of the attachment, let $P_1 \in \mathscr{P}$ be the leg, and let P_1 have ends $a_1 \in A$, $b_1 \in B$. Let P have ends u, v, let S, T be the switches with ends a_1, b_1, respectively, and let $\mathscr{S} = (\mathscr{P} - \{P_1\}) \cup \{S\}$, $\mathscr{T} = (\mathscr{P} - \{P_1\}) \cup \{T\}$. Now $u \notin V(\mathscr{P})$ and so u, a_1, b_1 are distinct. We assume without loss of generality that $u \in A$ and that (u, a_1, b_1) is clockwise. Let $W = V(\mathscr{P}) \cup \{u\}$. For $x \in \bar{\Omega}$, there is a unique element of W, denoted x^+, such that $x^+ \neq x$ and such that (x, x^+, y) is clockwise for all $y \in W - \{x, x^+\}$. We define x^- similarly. There is a unique $a \in A \cap W$ such that $a^- \notin A$, the "first" vertex of $A \cap W$. Similarly, let b be the "first" vertex of $B \cap W$ and let a', b' be the "last" vertices of $A \cap W$ and $B \cap W$, respectively. Several relationships exist amongst these vertices.

(1) *Either $u^- = b_1$, or $a_1 = a'$ and there exists $P_2 \in \mathscr{P} - \{P_1\}$ with ends a_1^+, a_1^-. In either case $a_1^- \neq u$.*

For $\mathscr{T}$ is certainly a transaction, with fence (A, B) and so it is not [p. 48]

crooked. Hence $a_1^- \neq u$. If T is peripheral in $\mathscr{T}$ then $u^- = b_1$, and if some $P_2 \in \mathscr{P} - \{P_1\}$ is peripheral in $\mathscr{T}$ then P_2 has ends a_1^+, a_1^- because P_2 is not peripheral in $\mathscr{P}$, and $a_1 = a'$ because (A, B) is a fence for $\mathscr{P}$.

(2) *Either $a \neq u$ or $b = b_1$.*

For let us assume that $a = u$. Then $\mathscr{S}$ is a transaction with fence $(A - \{a\}, B \cup \{a\})$. It is not crooked, and so $b = b_1$, since $a_1^- \neq u$ by (1).

(3) *Either $a' \neq a_1$ or there exists $P_3 \in \mathscr{P} - \{P_1\}$ with ends b_1^+, b_1^-.*

For let us assume that $a' = a_1$. Then $\mathscr{S}$ is a transaction with fence $(A - \{a'\}, B \cup \{a'\})$. It is not crooked, and by (1), $a_1^- \neq u$, so there exists $P_3 \in \mathscr{P} - \{P_1\}$ which is peripheral in $\mathscr{S}$ and not peripheral in $\mathscr{P}$. (3) follows.

Now to complete the proof, suppose that $u^- = b_1$. Then $a = u$, and so by (2), $b = b_1$. But this implies that $|B \cap W| = 1$ and so $|\mathscr{P}| = 1$, a contradiction. We deduce that $u^- \neq b_1$, and so by (1), $a_1 = a'$ and there exists $P_2 \in \mathscr{P} - \{P_1\}$ with ends a_1^+, a_1^-. By (3), there exists $P_3 \in \mathscr{P} - \{P_1\}$ with ends b_1^+, b_1^-. It follows that P_1 is an overpass of $\mathscr{P}$, as required. This completes the proof. □

We need a version of the exchange property which applies to overpass attachments, as follows.

(3.3) *Let $\mathscr{P}$ be a crooked transaction in (G, Ω). If P and P' are both non-exchangeable attachments to $\mathscr{P}$ and P, P' are disjoint, then there is a crooked transaction $\mathscr{P}'$ with $|\mathscr{P}'| = |\mathscr{P}| + 1$ and*

$$\bigcup \mathscr{P}' \subseteq \bigcup \mathscr{P} \cup (P \cup P').$$

Moreover, the two attachments have a common leg, which is an overpass of $\mathscr{P}$, and no edge of this leg between the two feet is in $\bigcup \mathscr{P}'$.

Proof. By (3.1) and (3.2) the two attachments have a common leg, P_1, say, and it is an overpass of $\mathscr{P}$. Let S be a switch of the first attachment and T a switch of the second, chosen so that no edge of P_1 between the two feet lies in S or T. Then S and T are disjoint, and it is easy to see that

$$\mathscr{P}' = (\mathscr{P} - \{P_1\}) \cup \{S, T\}$$

satisfies the theorem. □

We shall also need the following observation.

(3.4) *Let $p \geq 2$ be an integer, with $p \neq 3$. If $\mathscr{P}$ is a crooked transaction of cardinality $\geq p$ then some subset of $\mathscr{P}$ is a crooked transaction of cardinality p.*

The proof is easy.

We give the details. Let (A, B) be a fence for $\mathscr{P}$, and let $W = V(\mathscr{P})$. Since $\mathscr{P}$ is non-null, W meets both A and B. So, going once around Ω, we find $a' \in A \cap W$ and $b \in B \cap W$ with $\Omega(a', b) \cap W = \emptyset$, and in the same way $b' \in B \cap W$ and $a \in A \cap W$ with $\Omega(b', a) \cap W = \emptyset$.

(1) *Let $x \in A \cap W$ and $y \in B \cap W$. If $x \neq a'$ then $a' \in \Omega(x, y)$, and if $x = a'$ and $y \neq b$ then $b \in \Omega(x, y)$. Similarly, if $y \neq b'$ then $b' \in \Omega(y, x)$, and if $y = b'$ and $x \neq a$ then $a \in \Omega(y, x)$.*

For the first statement, let $x \neq a'$. Then $x \in \Omega(b, a')$, since $x \in W$ and $x \neq b$. If $y = b$ this says that $a' \in \Omega(x, y)$. If $y \neq b$ then also $y \in \Omega(b, a')$, and if (b, x, y, a') were clockwise then $\{x, a'\} \subseteq A$ would cross $\{b, y\} \subseteq B$, contrary to the definition of a fence. So (b, y, x, a') is clockwise, and again $a' \in \Omega(x, y)$. For the second statement, $y \in \Omega(b, a')$ since $y \in W$ and $y \neq b$, and so $b \in \Omega(a', y)$. The last two statements follow in the same way. This proves (1).

Suppose first that $p = 2$. Let $P_1, P_2 \in \mathscr{P}$ have ends a' and b respectively. If $P_1 = P_2$ then $\Omega(a', b) \cap W = \emptyset$ shows that P_1 is peripheral in $\mathscr{P}$, which is impossible. So $P_1 \neq P_2$, and we may write their ends as a', b_1 and a_2, b, where $b_1 \neq b$ and $a_2 \neq a'$, since P_1 and P_2 are disjoint. By (1), $a' \in \Omega(a_2, b_1)$ and $b \in \Omega(a', b_1)$, and so (a_2, a', b, b_1) is clockwise and $\{a', b_1\}$ crosses $\{a_2, b\}$. Hence each of $\Omega(a', b_1)$, $\Omega(b_1, a')$ contains an end of P_2, and each of $\Omega(a_2, b)$, $\Omega(b, a_2)$ contains an end of P_1. Thus $\{P_1, P_2\}$ is a crooked transaction, with fence (A, B).

Now suppose that $p \geq 4$. It is enough to show that if $|\mathscr{P}| \geq 5$ then $\mathscr{P} - \{P\}$ is crooked for some $P \in \mathscr{P}$, for then we may remove members one at a time until p of them remain. At most four members of $\mathscr{P}$ have one of a', b, b', a as an end, and so some $P \in \mathscr{P}$ has none of them as an end. Let $\mathscr{P}' = \mathscr{P} - \{P\}$, a transaction with fence (A, B). Then $V(\mathscr{P}')$ contains a', b, b', a. Let $R \in \mathscr{P}'$ have ends $x \in A$ and $y \in B$. Suppose that $\Omega(x, y) \cap V(\mathscr{P}') = \emptyset$. If $x \neq a'$ then $a' \in \Omega(x, y)$ by (1), a contradiction. If $x = a'$ then $y \neq b$, for otherwise $\Omega(x, y) \cap W = \emptyset$ and R would be peripheral in $\mathscr{P}$, and so $b \in \Omega(x, y)$ by (1), again a contradiction. In the same way, using b' and a, the set $\Omega(y, x) \cap V(\mathscr{P}')$ is not empty. So R is not peripheral in $\mathscr{P}'$, and $\mathscr{P}'$ is crooked.[F1]

[p. 49] 4. Separating Ω-Paths

Let (G, Ω) be a society. An Ω-path P is *non-separating* if there is an Ω-path P' such that P' crosses P, and *separating* otherwise. It is easy to

[F1] **Missing proof.** The paper says that the proof is easy and gives no proof. The argument above is the one used in the formalization.

see that P is separating if and only if there is a separation (G_1, G_2) of G with $V(P) = V(G_1 \cap G_2)$, $\Omega(s,t) \subseteq V(G_1)$, and $\Omega(t,s) \subseteq V(G_2)$, where P has ends s, t.

(4.1) *Let $\mathscr{P}$ be a transaction in (G, Ω), such that every peripheral member is non-separating. Then there is a crooked transaction $\mathscr{P}'$ in (G, Ω) with $|\mathscr{P}'| \geq |\mathscr{P}|$.*

Proof. Let the number of peripheral members of $\mathscr{P}$ be k. If $k = 0$ the result is true, and so we assume that $k = 1$ or 2. We shall prove there is a transaction with cardinality at least $|\mathscr{P}|$ and with $\leq k-1$ peripheral members, all non-separating, from which the result follows.

Let (A, B) be a fence for $\mathscr{P}$. Choose $P \in \mathscr{P}$, peripheral in $\mathscr{P}$, and let its ends be $a \in A$, $b \in B$. Without loss of generality we assume that $\Omega(a,b) \cap V(\mathscr{P}) = \emptyset$. Now P is non-separating, and so there is an Ω-path Q such that Q crosses P. If $Q \cap \bigcup \mathscr{P}$ is null then $\mathscr{P} \cup \{Q\}$ is a transaction satisfying our requirement; and so we assume that $Q \cap \bigcup \mathscr{P}$ is non-null. Let Q have ends s, t where $s \in \Omega(a,b)$, $t \in \Omega(b,a)$. Let Q' be a minimal subpath of Q between s and $\bigcup \mathscr{P}$. Then Q' is an attachment to $\mathscr{P}$. Let its leg be P_1, and let the corresponding switches be S, S', and let $\mathscr{P}' = (\mathscr{P} - \{P_1\}) \cup \{S\}$, $\mathscr{P}'' = \mathscr{P} - \{P_1\} \cup \{S'\}$. Both $\mathscr{P}'$ and $\mathscr{P}''$ are transactions, and so it suffices to show that in one of $\mathscr{P}', \mathscr{P}''$, every peripheral member is also peripheral in $\mathscr{P}$. Suppose not, and choose $P' \in \mathscr{P}'$ peripheral in $\mathscr{P}'$ but not in $\mathscr{P}$; and choose $P'' \in \mathscr{P}''$ similarly. Then clearly P, P', P'' are all distinct, and no member of $\mathscr{P} - \{P_1\}$ crosses P' or P''. Thus P, P', P'' are all peripheral in the transaction $\mathscr{P} - \{P_1\}$, which is impossible. This completes the proof. □

Let **S** be the following statement, which we extract here because it is a common hypothesis of several lemmas which follow.

Statement **S**. *$\mathscr{Q}$ is a transaction in a society (G, Ω), such that no member of $\mathscr{Q}$ has an internal vertex in $\bar{\Omega}$. A crooked transaction $\mathscr{P}$ in (G, Ω) is chosen with $|\mathscr{P}| \geq p$ (where $p \geq 2$ is some given integer), such that*

(i) *$\bigcup \mathscr{P} \cup \bigcup \mathscr{Q}$ is minimal*

(ii) *subject to (i), $|\mathscr{P}|$ is minimum,*

and (A, B) is a fence for $\mathscr{P}$.

If **S** is true, we define $W(\mathscr{Q})$ to be the set of those vertices $v \in V(\mathscr{Q})$ such that $v \notin V(\mathscr{P})$ and $Q \cap \bigcup \mathscr{P}$ is non-null, where Q is the member of $\mathscr{Q}$ with [p. 50] end v. If $v \in W(\mathscr{Q})$, we define the *attachment corresponding to* v to be the minimal subpath of $\bigcup \mathscr{Q}$ between v and $\bigcup \mathscr{P}$.

(4.2) *Let* **S** *be true. Suppose that $w, w' \in W(\mathscr{Q})$ are distinct, and that neither of the corresponding attachments is exchangeable. Then w, w' are*

both ends of the same member of $\mathscr{Q}$.

Proof. Let the corresponding attachments be P, P', with feet v, v', respectively. By (3.1) and (3.2), $\mathscr{P}$ has an overpass P'' and $v, v' \in V(P'')$. If $v = v'$ the result is true, and we assume $v \neq v'$, so that P and P' are disjoint. By (3.3) there is a crooked transaction $\mathscr{P}'$ with $|\mathscr{P}'| = |\mathscr{P}| + 1$ and

$$\bigcup \mathscr{P}' \subseteq \bigcup \mathscr{P} \cup (P \cup P'),$$

and such that no edge of P'' between v and v' lies in $\bigcup \mathscr{P}'$. From the choice of $\mathscr{P}$, all the edges of P'' between v and v' lie in $\bigcup \mathscr{Q}$, and the result follows. □

(4.3) *Let* **S** *be true. For every $Q \in \mathscr{Q}$ except possibly one, either*

(i) *$Q \cap \bigcup \mathscr{P} = \emptyset$, or*

(ii) *both ends of Q are in $V(\mathscr{P})$, or*

(iii) *Q meets exactly one member P of $\mathscr{P}$, $P \cap Q$ is a path, and P, Q and $P \cap Q$ have a common end, and the other end of Q is not in $V(\mathscr{P})$.*

If (i), (ii), (iii) are all false for some $Q \in \mathscr{Q}$ then $\mathscr{P}$ has an overpass.

Proof. We assume that $Q \in \mathscr{Q}$ satisfies none of (i), (ii), (iii). Thus Q meets $\bigcup \mathscr{P}$, and some end w of Q is not in $V(\mathscr{P})$. Hence $w \in W(\mathscr{Q})$; let P be the corresponding attachment, with leg P_1 and foot s. Suppose that it is exchangeable. Then from the minimality of $\bigcup \mathscr{P} \cup \bigcup \mathscr{Q}$ it follows that some subpath of P_1 from s to an end t of P_1 lies completely in $\bigcup \mathscr{Q}$ and hence in Q. Then $t \in V(Q)$, and since no internal vertex of Q is in $\bar{\Omega}$ we deduce that t is the end of Q different from w. Hence (iii) is true, a contradiction. It follows that the corresponding attachment is not exchangeable, and by (3.2), that $\mathscr{P}$ has an overpass. Now if $Q' \in \mathscr{Q}$ also satisfies none of (i), (ii), (iii), then $Q = Q'$, by (4.2). The result follows. □

If Q satisfies (4.3) (iii) we say that Q is *half-covered*.

Let $P_1, ..., P_k$ be mutually disjoint Ω-paths. We say that $P_1, ..., P_k$ are *in order* if there is a clockwise sequence

$$(a_k, a_{k-1}, ..., a_2, a_1, b_1, b_2, ..., b_{k-1}, b_k),$$

where P_i has ends a_i, b_i $(1 \leq i \leq k)$. It follows that no two of $P_1, ..., P_k$ cross.

(4.4) *Let* **S** *be true, and let $Q_1, Q_2, Q_3 \in \mathscr{Q}$ be in order, where Q_2 is* [p. 51] separating. Suppse that Q_1, Q_2, Q_3 are all half-covered. Then one end of Q_2 is in A and the other is in B.*

Proof. Let Q_i have ends a_i, b_i $(1 \leq i \leq 3)$, where $(a_3, a_2, a_1, b_1, b_2, b_3)$ is clockwise. Suppose that $a_2, b_2 \in A$. Then either $B \subseteq \Omega(a_2, b_2)$ or $B \subseteq$

$\Omega(b_2, a_2)$; we assume the second without loss of generality. Choose $P_i \in \mathscr{P}$ such that P_i meets Q_i $(i = 1, 2, 3)$. Now both ends of Q_1 are in A, but one end of P_1 is in B (and the other is one of a_1, b_1). Q_2 is separating and so P_1 does not cross Q_2; hence $P_1 \cap Q_2 \neq \emptyset$. But Q_2 only meets one member of $\mathscr{P}$, namely, P_2, and so $P_1 = P_2$. It follows that one of a_2, b_2 is an end of P_1, and so both ends of P_1 are in A, a contradiction. Thus not both a_2, b_2 are in A, and similarly they are not both in B, as required. □

(4.5) *Let S be true, and let $Q_1, Q_2, Q_3, Q_4, Q_5 \in \mathscr{Q}$ be in order. Suppose that Q_2, Q_3, Q_4 are all separating. Then not all of $Q_1, ..., Q_5$ are half-covered.*

Proof. Suppose that $Q_1, ..., Q_5$ are all half-covered. For $1 \leq i \leq 5$, choose $P_i \in \mathscr{P}$ so that Q_i meets P_i. Let Q_i have ends a_i, b_i $(1 \leq i \leq 5)$ where $(a_5, a_4, ..., a_1, b_1, ..., b_5)$ is clockwise. By (4.4) we may assume that $a_2, a_3, a_4 \in A$ and $b_2, b_3, b_4 \in B$. In particular, $\mathscr{P}' = (\mathscr{P} - \{P_3\}) \cup \{Q_3\}$ is a transaction, and so it is not crooked, from the minimality of $\bigcup \mathscr{P} \cup \bigcup \mathscr{Q}$. Thus either Q_3 is peripheral in $\mathscr{P}'$, or some $P \in \mathscr{P}' - \{Q_3\}$ is peripheral in $\mathscr{P}'$.

Suppose that Q_3 is peripheral in $\mathscr{P}'$. We assume without loss of generality that $V(\mathscr{P}') \cap \Omega(a_3, b_3) = \emptyset$. Now both P_1 and P_2 have an end in $\Omega(a_3, b_3)$, and so neither P_1 nor P_2 is in $\mathscr{P}'$. Hence $P_1 = P_2 = P_3$, which is impossible since P_i and Q_i have a common end $(i = 1, 2, 3)$.

We deduce that some $P \in \mathscr{P}' - \{Q_3\}$ is peripheral in $\mathscr{P}'$. Let the ends of P be a, b where $a \in A$, $b \in B$. Now Q_3 does not cross P, and so we may assume without loss of generality that (a_3, a, b, b_3) is clockwise. Thus $V(\mathscr{P}') \cap \Omega(a, b) = \emptyset$. Let the ends of P_3 be a_3, b_3', say (again, without loss of generality), where $b_3' \in B$. Then $b_3' \in \Omega(a, b)$ since P is not peripheral in $\mathscr{P}$.

Suppose that $b_2 \notin \Omega(a, b)$. Then $b_2 \in \Omega(b_3', b_3)$ since $b_2 \in B$, and $a_2 \in \Omega(a_3, b_3')$ since $a_2 \in A$. But P_3 does not cross Q_2 and so $Q_2 \cap P_3 \neq \emptyset$; hence $P_2 = P_3$, and b_3' is an end of Q_2, which (since $b_3' \in B$ and $a_2 \in A$) implies that $b_2 = b_3' \in \Omega(a, b)$, contrary to our assumption.

We deduce that $b_2 \in \Omega(a, b)$. If $a_2 \in \Omega(a_3, a)$ then $P \cap Q_2 \neq \emptyset$ since P does not cross Q_2, and so $P = P_2$, which implies that $\{a, b\} \cap \{a_2, b_2\} \neq \emptyset$, a contradiction. We deduce that $a_2 \notin \Omega(a_3, a)$, and so $a_2 \in \{a\} \cup \Omega(a, b_3')$, since $a_2 \in A$. Since Q_1, Q_2, Q_3 are in order, it follows that

$$a_1, b_1 \in \Omega(a_2, b_2) \subseteq \Omega(a, b)$$

[p. 52] and so P_1 has an end in $\Omega(a, b)$. But $V(\mathcal{P}') \cap \Omega(a, b) = \emptyset$, and so $P_1 \notin \mathcal{P}'$, and hence $P_1 = P_3$. Now P_1 meets Q_1 and P_3 meets Q_3, and Q_1, Q_2, Q_3 are in order, and so $P_1 = P_3$ meets or crosses Q_2. The second is

impossible since Q_2 is separating, and so $P_1 = P_2 = P_3$, which is impossible as before. This completes the proof. □

(4.6) *Let S be true, with $p \geq 2$ and $p \neq 3$, let $|\mathcal{Q}| = p + 4$, and let every member of $\mathcal{Q}$ be separating. Then $Q \cap \bigcup \mathcal{P}$ is null for some $Q \in \mathcal{Q}$.*

Proof. Suppose each $Q \in \mathcal{Q}$ meets $\bigcup \mathcal{P}$. By (4.3), for each $Q \in \mathcal{Q}$ either Q is half-covered or both ends of Q are in $V(\mathcal{P})$, with at most one exception. By (4.5) there are at most four members of $\mathcal{Q}$ which are half-covered. We deduce that

$$|V(\mathcal{P})| \geq 2(|\mathcal{Q}| - 5) + 4 = 2p + 2.$$

Hence $|\mathcal{P}| > p$ which by (3.4) contradicts the minimality of $|\mathcal{P}|$. □

5. Separations And Crosses

We continue with lemmas for the main proof.

(5.1) *Let Q be a separating Ω-path in (G, Ω) with ends a, b. If $\mathcal{P}$ is a crooked transaction in (G, Ω) with $Q \cap \bigcup \mathcal{P}$ null and with $V(\mathcal{P}) \cap \Omega(a, b) \neq \emptyset$ then there is a cross $\mathcal{P}' \subseteq \mathcal{P}$ with all four ends in $\Omega(a, b)$.*

Proof. If $P \in \mathcal{P}$ has an end in $\Omega(a, b)$, then both its ends are in $\Omega(a, b)$ since P does not cross Q. Let us choose $P \in \mathcal{P}$ with ends $s, t \in \Omega(a, b)$, where (a, s, t, b) is clockwise, such that $\Omega(s, t)$ is minimal. Since P is not peripheral in $\mathcal{P}$, there exists $P' \in \mathcal{P} - \{P\}$ with an end in $\Omega(s, t)$. Its other end is in $\Omega(a, b)$ but not in $\Omega(s, t)$, from the minimality of $\Omega(s, t)$. Then $\{P, P'\}$ is the required cross. □

(5.2) *Let (G, Ω) have a crooked transaction with cardinality $p \geq 2$ and $p \neq 3$. Let $\{Q_1, \ldots, Q_{p+4}\}$ be a transaction in (G, Ω) such that for $1 \leq i \leq p + 4$, Q_i is separating, and has no internal vertex in $\bar{\Omega}$, and has ends a_i, b_i, where $(a_{p+4}, a_{p+3}, \ldots, a_1, b_1, b_2, \ldots, b_{p+4})$ is clockwise. Let (G_1, G_2) be a separation of G with $V(G_1 \cap G_2) = V(Q_{p+4})$, $\Omega(a_{p+4}, b_{p+4}) \subseteq V(G_1)$ and $\Omega(b_{p+4}, a_{p+4}) \subseteq V(G_2)$. Then either*

(i) *there is a cross $\{P_1, P_2\}$ of (G, Ω) with $P_1 \cap G_2$ and $P_2 \cap G_2$ both null, or*

(ii) *there is a crooked transaction $\mathcal{P}$ in (G, Ω) with cardinality p and with $\Omega(a_1, b_1) \cap V(\mathcal{P}) = \emptyset$.*

[p. 53] *Proof.* Let $\mathcal{Q} = \{Q_1, \ldots, Q_{p+4}\}$, and choose a crooked transaction $\mathcal{P}$ in (G, Ω) such that the statement S of the previous section is true. By (3.4), $|\mathcal{P}| = p$, and by (4.6), $Q_j \cap \bigcup \mathcal{P}$ is null for some j $(1 \leq j \leq p + 4)$. We assume that (ii) does not hold, and so $V(\mathcal{P}) \cap \Omega(a_j, b_j) \neq \emptyset$. Hence by (5.1) there is a cross $\{P_1, P_2\} \subseteq \mathcal{P}$, with all four ends in $\Omega(a_j, b_j)$. Since Q_j is

separating it follows that $P_1 \cap G_2$ and $P_2 \cap G_2$ are both null, and (i) holds, as required. □

(5.3) *Let Q be an induced Ω-path in (K, Ω) with ends a, b such that $\Omega(a, b) = \emptyset$ and $\bar{\Omega} - \{a, b\}$ catches $K \setminus V(Q)$, and no internal vertex of Q is in $\bar{\Omega}$. Let $r \in V(Q) - \{a, b\}$, and let Ω_1 be the cyclic permutation in $V(K)$ with $\bar{\Omega}_1 = \bar{\Omega} \cup \{r\}$ and $\Omega_1 \mid \bar{\Omega} = \Omega$, and with (a, r, b) Ω_1-clockwise. If (K, Ω_1) has a cross then (K, Ω) has a cross or tripod.*

Proof. Let $\{P_1, P_2\}$ be a cross in (K, Ω_1), chosen with $P_1 \cup P_2 \cup Q$ minimal. Let P_i have ends s_i, t_i $(i = 1, 2)$. We may assume that $\{P_1, P_2\}$ is not a cross of (K, Ω), and so $r \in \{s_1, t_1, s_2, t_2\}$.

(1) *Either (K, Ω) has a cross or $a, b \in \{s_1, t_1, s_2, t_2\}$.*

For certainly Q meets $P_1 \cup P_2$; let A be a minimal subpath of Q from a to $V(P_1 \cup P_2)$, with ends a, a' where $a' \in V(P_1)$, say. Let S, T be the two subpaths of P_1 from a' to s_1, t_1, respectively. One of $\{A \cup S, P_2\}$, $\{A \cup T, P_2\}$ is a cross in (K, Ω_1) and we assume the former without loss of generality. From the minimality of $P_1 \cup P_2 \cup Q$ we deduce that $T \subseteq Q$ and in particular, $t_1 \in V(Q) \cap \bar{\Omega}_1 = \{a, b, r\}$. We may assume that $(A \cup S, P_2)$ is not a cross in (K, Ω) and so r is an end of one of $A \cup S, P_2$; and it follows that $r \notin V(T)$, since $P_1 \neq T$. Since $t_1 \in \{a, b, r\}$ and $r \notin V(T)$ we deduce that $a = t_1 \in \{s_1, t_1, s_2, t_2\}$, and similarly $b \in \{s_1, t_1, s_2, t_2\}$, as claimed.

Since $r \in \{s_1, t_1, s_2, t_2\}$, we may assume that $r = s_2$ without loss of generality. Then $t_2 \in \Omega_1(b, a)$, since $\Omega_1(a, b) = \{r\}$ and P_1 crosses P_2. From (1), we may assume that $s_1 = a$ and $t_1 = b$. Let R_2 be the minimal subpath of P_2 between $\{t_2\}$ and $V(Q)$, and let R_2 have ends t_2, c, say, where $c \in V(Q)$. Let C be the subpath of Q between a and c, and let D be the subpath between c and b. Now $a, b \in V(P_1)$, and so P_1 meets both C and D. Choose a minimal subpath R_1 of P_1 between $V(C)$ and $V(D)$. Now R_1 has more than one edge, since Q is induced, and so $V(R_1) \not\subseteq V(Q)$. But $\bar{\Omega} - \{a, b\}$ catches $K \setminus V(Q)$, and so there is a $V(R_1) \to (\bar{\Omega} - \{a, b\})$ path $P \subseteq K$ which avoids $V(Q)$. If $P \cap R_2$ is null then (K, Ω) has a cross $\{M_1, M_2\}$ where M_1 is the union of P, part of R_1 and part of Q, and M_2 is the union of R_2 and part of Q. If $P \cap R_2$ is non-null then (K, Ω) has a tripod $\{N_1, N_2, N_3\}$ the "ends" of which are c and the vertex of $P \cap R_1$, where $N_1, N_2 \subseteq Q \cup R_1$ and $N_3 \subseteq P \cup R_2$. This completes the proof. □

We shall also need the following extension of (2.3).

[p. 54] (5.4) *Let (G, Ω) be internally 4-connected. Let Q be a separating Ω-path with ends a, b and with no internal vertex in $\bar{\Omega}$, and let (G_1, G_2) be a separation of G such that $V(G_1 \cap G_2) = V(Q)$, $\Omega(a, b) \subseteq V(G_1)$ and $\Omega(b, a) \subseteq V(G_2)$. Let $G' = G \setminus V(G_2)$, let $\Omega' = \Omega \mid V(G')$, and let $\Omega_1 = \Omega \mid$*

$V(G_1)$. *If* (G', Ω') *has a tripod then* (G_1, Ω_1) *has a cross.*

Proof. Let G'' be the graph obtained from G_1 by contracting all the edges of Q, and let q be the new vertex thus formed. Let Ω'' be the cyclic permutation with $\bar{\Omega}'' = \bar{\Omega}' \cup \{q\}$ and $\Omega'' \mid V(G') = \Omega'$, such that (x, y, q) is Ω''-clockwise whenever (x, y, a) is Ω-clockwise and $x, y \in \bar{\Omega}'$.

Now if (G'', Ω'') has a cross then so does (G_1, Ω_1). We assume then that (G'', Ω'') has no cross. Let $\{P_1, P_2, P_3\}$ be a tripod of (G', Ω'); then it is also a tripod of (G'', Ω'') not meeting q. By (2.3), (G'', Ω'') has an internal 3-separation (G_1'', G_2'') with $P_1 \cup P_2 \cup P_3 \subseteq G_1''$. Then $(G_1'' \setminus q, G_2'' \setminus q)$ is an internal 3-separation of (G', Ω') with $P_1 \cup P_2 \cup P_3 \subseteq G_1'' \setminus q$, and yet $\{P_1, P_2, P_3\}$ is a tripod of (G', Ω'). Thus $|V((G_1'' \cap G_2'') \setminus q)| = 3$. Hence $q \notin V(G_1'' \cap G_2'')$, and it follows that (G, Ω) has an internal 3-separation, contrary to hypothesis. This completes the proof. □

We combine these lemmas for the following.

(5.5) *Let* (G, Ω) *be internally 4-connected and let* $p \geq 2$ *with* $p \neq 3$. *Let* $\{Q_1, \ldots, Q_{p+5}\}$ *be a transaction such that for* $1 \leq i \leq p+5$, Q_i *is separating, has no internal vertex in* $\bar{\Omega}$, *and has ends* a_i, b_i, *where* $(a_{p+5}, a_{p+4}, \ldots, a_1, b_1, b_2, \ldots, b_{p+5})$ *is clockwise. Let* $\Omega(a_1, b_1) = \emptyset$, *and let* $r \in V(Q_1) - \{a_1, b_1\}$. *Let* (G_1, G_2) *be a separation of* G *with* $V(G_1 \cap G_2) = V(Q_{p+5})$, $\Omega(a_{p+5}, b_{p+5}) \subseteq V(G_1)$, *and* $\Omega(b_{p+5}, a_{p+5}) \subseteq V(G_2)$. *Let* Ω_0 *be the cyclic permutation with* $\bar{\Omega}_0 = \bar{\Omega} \cup \{r\}$ *and* $\Omega_0 \mid \bar{\Omega} = \Omega$, *such that* (a_1, r, b_1) *is* Ω_0*-clockwise. Suppose that* Q_1 *is induced, that* $\bar{\Omega} - \{a_1, b_1\}$ *catches* $G \setminus V(Q_1)$, *that* (G, Ω_0) *has a crooked transaction of cardinality* p *and that* (G, Ω) *has no crooked transaction of cardinality* p. *Then there is a non-separating* Ω*-path* P *of* (G, Ω) *with* $P \subseteq G_1$.

Proof. We observe first that

(1) $Q_2, \ldots, Q_{p+5}$ *are separating* Ω_0*-paths of* (G, Ω_0).

For if $\{P, Q_j\}$ is a cross in (G, Ω_0) for some j with $2 \leq j \leq p+5$ then r is an end of P since Q_j is separating in (G, Ω); and we can choose an Ω-path $P' \subseteq P \cup Q_1$ with an end in $\Omega(b_j, a_j)$. But then $\{P', Q_j\}$ is a cross in (G, Ω), a contradiction.

Let K be the maximal subgraph of $G_1 \setminus V(Q_{p+5})$ caught by $\bar{\Omega} - V(G_2)$.

(2) *There is a cross in* $(K, \Omega_0 \mid V(K))$.

For by (1), we may apply (5.2) to the transaction $\{Q_2, \ldots, Q_{p+5}\}$ of (G, Ω_0). Certainly there is no crooked transaction $\mathcal{P}$ in (G, Ω_0) with [p. 55] cardinality p and with $\Omega_0(a_2, b_2) \cap V(\mathcal{P}) = \emptyset$, for $\mathcal{P}$ would be a crooked transaction in (G, Ω), since $r \in \Omega_0(a_2, b_2)$. Thus by (5.2), (2) follows.

(3) $\bar{\Omega} \cap V(K \setminus V(Q_1))$ *catches* $K \setminus V(Q_1)$.

For let H be a component of $K \setminus V(Q_1)$. We must show that

$$V(H) \cap \bar{\Omega} \cap V(K \setminus V(Q_1)) \neq \emptyset. \qquad (*)$$

Now since $H \subseteq K$ there is a path of K from $V(H)$ to $\bar{\Omega} - V(G_2) = \bar{\Omega} \cap V(K)$. If this path avoids $V(Q_1)$, then $(*)$ holds, and so we may assume that some vertex of H is adjacent in G to a vertex of Q_1. Now $\bar{\Omega} - \{a_1, b_1\}$ catches $G \setminus V(Q_1)$, and so there is a path of G from $V(H)$ to $\bar{\Omega} - \{a_1, b_1\}$ avoiding $V(Q_1)$. If this path avoids $V(G_2)$ then $(*)$ holds, and so we may assume that some vertex of H is adjacent in G to a vertex of Q_{p+5}. Now Q_2 is separating in (G, Ω), and H is connected, and there are vertices of Q_1 and of Q_{p+5} with neighbors in H; and so $V(H \cap Q_2) \neq \emptyset$. But then $Q_2 \subseteq H$ and $a_2, b_2 \in V(H)$, whence $(*)$ holds.

The proof is completed as follows. From (2), (3), and (5.3) applied to $(K, \Omega \mid V(K))$ we deduce that $(K, \Omega \mid V(K))$ has a cross or tripod, and hence so does $(G \setminus V(G_2), \Omega \mid (\bar{\Omega} - V(G_2)))$. Using (5.4) in the tripod case, we deduce that $(G_1, \Omega \mid V(G_1))$ has a cross, as required. □

6. The Main Theorem

In this section we apply (5.5) to deduce our main result. Let $\mathcal{P}$ be a transaction in (G, Ω), with $|\mathcal{P}| \geq 2$. If $P \in \mathcal{P}$ is peripheral and has ends s, t, we define $U(P) = \Omega(t, s)$, where $\Omega(s, t) \cap V(\mathcal{P}) = \emptyset$. (This defines $U(P)$ uniquely.) If $\mathcal{P}$ is crooked, we define $U(\mathcal{P}) = \bar{\Omega}$, and otherwise we define $U(\mathcal{P})$ to be the intersection of the sets $U(P)$, taken over all peripheral $P \in \mathcal{P}$. We define $C(\mathcal{P})$ to be the set of all vertices v of G such that there is a path in G between $\{v\}$ and $U(\mathcal{P})$ which avoids every peripheral member of $\mathcal{P}$.

Our main theorem is the following.

(6.1) *Let $p \geq 2$ be an integer with $p \neq 3$, and let (G, Ω) be a 3-connected, internally 4-connected society, in which there is no crooked transaction of cardinality p. Then the sophistication of (G, Ω) is at most $3p + 7$.*

Proof. We suppose for a contradiction that the theorem is false. Choose a counterexample (G, Ω) such that $|V(G)| + |E(G)|$ is as small as possible, and subject to this, $|\bar{\Omega}|$ is as large as possible. We remark that:

(1) *G is simple.*

[p. 56] For if e is a loop or multiple edge, then $(G \setminus e, \Omega)$ is a smaller counterexample.

(2) *No vertex $v \in \bar{\Omega}$ is isolated.*

For otherwise $(G \setminus v, \Omega \mid (\bar{\Omega} - \{v\}))$ is a smaller counterexample.

(3) *If $v \in \bar{\Omega}$ has valency 1, then its neighbor is in $\bar{\Omega}$.*

For otherwise (G', Ω) is a smaller counterexample, where G' is the graph obtained from G by contracting the edge incident with v (and Ω is interpreted as a cyclic permutation in $V(G')$ in the natural way).

(4) *If $u, v \in \bar{\Omega}$ are distinct and $\Omega(u, v) = \emptyset$, then u and v are not adjacent in G.*

For if an edge e joins u and v, then $(G \backslash e, \Omega)$ is a smaller counterexample. We claim that we may choose a transaction $\mathcal{P}$ in (G, Ω) such that

(a) $|\mathcal{P}| = 3p + 8$

(b) for each $P \in \mathcal{P}$, no internal vertex of P is in $\bar{\Omega}$, and

(c) subject to (a) and (b), $C(\mathcal{P})$ is maximal.

For it is possible to choose a transaction $\mathcal{P}$ satisfying (a) since (G, Ω) has sophistication $\geq 3p + 8$. Let (A, B) be a fence for $\mathcal{P}$, for each $P \in \mathcal{P}$ choose an $A \to B$ subpath of P, and let $\mathcal{P}'$ be the collection of these subpaths. Then $\mathcal{P}'$ satisfies (a) and (b); and so the choice of $\mathcal{P}$ satisfying (a), (b), and (c) is possible.

We assume now that $P \in \mathcal{P}$ is peripheral, with ends a, b where $\Omega(a, b) \cap V(\mathcal{P}) = \emptyset$, and that P is separating. Let (G_1, G_2) be a separation of G with $V(P) = V(G_1 \cap G_2)$, $\Omega(a, b) \subseteq V(G_1)$ and $\Omega(b, a) \subseteq V(G_2)$, chosen so that G_2 is minimal. Under these conditions the following remarks (5)-(8) apply.

(5) *If P' is an Ω-path with $P' \subseteq G_1$, then $C(\mathcal{P}) \subseteq C((\mathcal{P} - \{P\}) \cup \{P'\})$.*

We observe first that $\mathcal{P}' = (\mathcal{P} - \{P\}) \cup \{P'\}$ is a transaction; and certainly $U(\mathcal{P}) \subseteq U(\mathcal{P}')$. Now suppose that $v \in C(\mathcal{P})$, and let Q be a path of G between $\{v\}$ and $U(\mathcal{P})$ disjoint from every peripheral member of $\mathcal{P}$. Then in particular $Q \cap P$ is null, and so $Q \subseteq G_2 \setminus V(G_1)$. Hence $Q \cap P'$ is null, and so Q is disjoint from every peripheral member of $\mathcal{P}'$, which implies that $v \in C(\mathcal{P}')$. Hence $C(\mathcal{P}) \subseteq C(\mathcal{P}')$.

(6) *If P' is an Ω-path with $P' \subseteq G_1$, then P' has ends a, b.*

Suppose that P' has an end $c \neq a, b$. Choose a minimal Ω-path $P'' \subseteq P'$ with c one end. Then no internal vertex of P'' is in $\bar{\Omega}$. Let $\mathcal{P}' = (\mathcal{P} - \{P\}) \cup \{P''\}$. Then $\mathcal{P}'$ satisfies (a) and (b), and by (5), $C(\mathcal{P}) \subseteq C(\mathcal{P}')$; hence by (c), $C(\mathcal{P}) = C(\mathcal{P}')$. However, one end of P is in $U(\mathcal{P}') \subseteq C(\mathcal{P}')$, and neither end of P is in $C(\mathcal{P})$. This is a contradiction.

[p. 57] (7) $\Omega(a, b) = \emptyset$.

For suppose that $c \in \Omega(a, b)$. Let G' be the component of G containing c. If $G' \cap P$ is not null, let P_1 be a $\{c\} \to V(G' \cap P)$ path in G'; then $P_1 \subseteq G_1$, since no internal vertex of P_1 is in $G_1 \cap G_2$ and $c \in V(G_1) - V(G_2)$. Hence there is an Ω-path P' with $P' \subseteq G_1$ and one end c, contrary to (6). Thus $G' \cap P$ is null, and so $G' \subseteq G_1$. By (6), there is no Ω-path P' with $P' \subseteq G'$, and so only one vertex of G' is in $\bar{\Omega}$. But (G, Ω) is 3-connected, and so $V(G') = \{c\}$, contrary to (2). This verifies (7).

Let $H = G_2 \setminus V(G_1)$. By the minimality of G_2, $\Omega(b, a)$ catches H, and G_1 is an induced subgraph of G.

(8) *If H' is a subgraph of G such that $H \subseteq H'$, $\Omega(b,a)$ catches H', and there is a path of G between a and b avoiding $V(H')$, then $H' = H$.*

For let P' be a path of G between a and b avoiding $V(H')$. Then $P' \subseteq G_1$, since $V(P') \subseteq V(G_1)$ and G_1 is an induced subgraph of G. Let $\mathcal{P}' = (\mathcal{P} - \{P\}) \cup \{P'\}$; then $\mathcal{P}'$ is a transaction satisfying (a) and (b) (since $\Omega(b,a) \subseteq V(H)$ and $\Omega(a,b) = \emptyset$). By (5), $C(\mathcal{P}) \subseteq C(\mathcal{P}')$, and so by (c), $C(\mathcal{P}) = C(\mathcal{P}')$. Suppose that $H \neq H'$. Now H is an induced subgraph of G, and so there is a vertex v of H' which is not a vertex of H. Since $v \in V(H')$ and $\Omega(b,a)$ catches H', there is a $\{v\} \to \Omega(b,a)$ path Q which avoids $V(P')$. Since $v \notin V(H)$ we have $v \in V(G_1)$, and so $P \cap Q$ is non-null; and in particular it follows that we may choose v and Q so that $v \in V(P)$ and Q is a $V(P) \to \Omega(b,a)$ path. Now $v \in V(P)$ and so $v \notin C(\mathcal{P}) = C(\mathcal{P}')$. We deduce that $V(H') \not\subseteq C(\mathcal{P}')$ and so $\mathcal{P}'$ has exactly one other peripheral member $P'' \neq P'$. Let Q' be the minimal subpath of Q between $\{v\}$ and $V(P'') \cup \Omega(b,a)$, and let v, u be the ends of Q'. Then Q' meets no non-peripheral member of $\mathcal{P}'$, because $v \notin C(\mathcal{P}')$. Let P'' have ends a'', b'' where (a, b, b'', a'') is clockwise. Then $u \notin \Omega(a'', b'')$, because $v \notin C(\mathcal{P}')$. Thus either $u \in \Omega(b'', a'')$ or $u \in V(P'')$. Hence any $p-1$ members of $\mathcal{P} - \{P, P''\}$, together with a suitable path R with $Q' \subseteq R \subseteq P \cup Q' \cup P''$, constitute a crooked transaction with cardinality p in (G, Ω), which is a contradiction. We deduce that $H' = H$, as claimed. This proves (8).

As $H = G_2 \setminus V(G_1)$ and G_1 is induced, it follows that $G_1 = G \setminus V(H)$. Hence, if $G_1 \neq P$, then by (2.1) there would be a separation (J_1, J_2) of G with $\{a, b\} \subseteq V(J_1)$, $|V(J_1 \cap J_2)| = 2$, $H \subseteq J_1$ and $V(J_1) \neq V(G)$. But then (J_2, J_1) would by (7) be a separation of (G, Ω), contradicting the 3-connectedness of (G, Ω). It follows that $G_1 = P$.

These results can be summarized as follows.

(9) *If $P \in \mathcal{P}$ is peripheral and separating, with ends a, b where $\Omega(a,b) \cap V(\mathcal{P}) = \emptyset$, then $\Omega(a,b) = \emptyset$, P is induced, and $\Omega(b,a)$ catches $G \setminus V(P)$.*

[p. 58] For an integer t with $0 \leq t \leq 2$, let $\mathbf{S}_t$ be the following statement.

Statement $\mathbf{S}_t$. The society (G, Ω) has a transaction $\mathcal{P}_t$ with $|\mathcal{P}_t| \geq 3p + 8 - t(p+4)$ and with at most $2 - t$ peripheral separating members. Moreover, every separating member of $\mathcal{P}_t$ is in $\mathcal{P}$, and every peripheral separating member of $\mathcal{P}_t$ is peripheral in $\mathcal{P}$.

Now $\mathbf{S}_0$ is true, with $\mathcal{P}_0 = \mathcal{P}$. We shall show that for $t = 0$ or $t = 1$, if $\mathbf{S}_t$ is true then $\mathbf{S}_{t+1}$ is true.

Put $|\mathcal{P}_t| = n$, and let $\mathcal{P}_t = \{P_1, \ldots, P_n\}$; let P_i have ends $a_i \in A$, $b_i \in B$ $(1 \leq i \leq n)$, where (A, B) is a fence for $\mathcal{P}_t$, and choose this lettering so that $(a_n, a_{n-1}, \ldots, a_1, b_1)$ is clockwise. Now if $\mathcal{P}_t$ has no peripheral separating members then we may set $\mathcal{P}_{t+1} = \mathcal{P}_t$ and so $\mathbf{S}_{t+1}$ is true. We assume then

without loss of generality that P_1 is peripheral in $\mathcal{P}_t$ and is separating. Thus $P_1 \in \mathcal{P}$ and P_1 is peripheral in $\mathcal{P}$. If one of $P_2, \ldots, P_{p+5}$ is non-separating (say P_j), then we may set $\mathcal{P}_{t+1}$ to be $\{P_j, P_{p+6}, P_{p+7}, \ldots, P_n\}$. We assume then that $P_1, \ldots, P_{p+5}$ are all separating and hence all in $\mathcal{P}$. It follows that $P_1, \ldots, P_{p+5}$ are in order and that, by (b), none of them has an internal vertex in $\bar{\Omega}$.

Now from $\mathbf{S}_t$, P_1 is peripheral in $\mathcal{P}$, and hence from (9), $\Omega(a_1, b_1) = \emptyset$. By (4), a_1 and b_1 are not adjacent, and so we may choose $r \in V(P_1)$ with $r \neq a_1, b_1$. Let Ω_0 be the cyclic permutation in $V(G)$ with $\bar{\Omega}_0 = \bar{\Omega} \cup \{r\}$ and $\Omega_0 \mid \bar{\Omega} = \Omega$, such that (a_1, r, b_1) is Ω_0-clockwise. Then $|\bar{\Omega}_0| > |\bar{\Omega}|$, and so from the maximality of $|\bar{\Omega}|$ the theorem is true for (G, Ω_0). But (G, Ω) has sophistication $\geq 3p + 8$ and hence so does (G, Ω_0). Therefore (G, Ω_0) has a crooked transaction of cardinality p.

Let (G_1, G_2) be a separation of G with $V(G_1 \cap G_2) = V(P_{p+5})$, $\Omega(a_{p+5}, b_{p+5}) \subseteq V(G_1)$, and $\Omega(b_{p+5}, a_{p+5}) \subseteq V(G_2)$. Then $P_{p+6}, \ldots, P_n$ are disjoint from G_1. By applying (5.5) to the transaction $\{P_1, \ldots, P_{p+5}\}$ in (G, Ω) (which is possible by (9)) we deduce that there is a non-separating path P' of (G, Ω) with $P' \subseteq G_1$. Thus we may set $\mathcal{P}_{t+1} = \{P', P_{p+6}, \ldots, P_n\}$ and $\mathbf{S}_{t+1}$ is true. This completes the proof that $\mathbf{S}_{t+1}$ is true.

We deduce that $\mathbf{S}_2$ is true. But every peripheral member of $\mathcal{P}_2$ is non-separating and $|\mathcal{P}_2| \geq p$, and so (G, Ω) has a crooked transaction with cardinality p, by (4.1) and (3.4). This is a contradiction, and completes the proof of the theorem. □

7. Some Special Transactions

The remainder of the paper is devoted to proving certain lemmas which will make the application of (6.1) easier in future papers. (6.1) tells us (roughly) that either (G, Ω) has sophistication at most $3p + 7$ or it has a crooked transaction of cardinality p. In applications it is useful to know [p. 59] more about the structure of the transaction than just that it is crooked, even at some sacrifice of size. In this section we prove that every large crooked transaction contains a (fairly) large crooked transaction which is of one of three special types. We now describe these three types.

Let $\mathcal{P}$ be a transaction. If there is a fence (A, B) for $\mathcal{P}$, and an enumeration $\mathcal{P} = \{P_1, \ldots, P_p\}$, where P_i has ends $a_i \in A$, $b_i \in B$ $(1 \leq i \leq p)$, such that $(a_p, a_{p-1}, \ldots, a_1, b_1)$ is clockwise, and

(i) $p \geq 2$ and $(a_p, a_{p-1}, \ldots, a_2, a_1, b_p, b_{p-1}, \ldots, b_2, b_1)$ is clockwise, we say $\mathcal{P}$ is a *crosscap*

(ii) $p \geq 3$ and $(a_p, a_{p-1}, \ldots, a_2, a_1, b_2, b_3, \ldots, b_p, b_1)$ is clockwise, we

say $\mathcal{P}$ is a *leap*

(iii) $p \geq 4$ and $(a_p, a_{p-1}, \ldots, a_2, a_1, b_2, b_1, b_3, b_4, \ldots, b_{p-3}, b_{p-2}, b_p, b_{p-1})$ is clockwise, we say $\mathcal{P}$ is a *doublecross*.

These are represented pictorially in Fig. 3.

The result of this section is the following.

(7.1) *Let $r \geq 2$, $s \geq 3$, $t \geq 4$ be integers. Let $\mathcal{P}$ be a crooked transaction in (G, Ω) such that $|\mathcal{P}| \geq (r-1)(2s+t-7)$. Then there exists $\mathcal{P}' \subseteq \mathcal{P}$ such that either*

(i) *$|\mathcal{P}'| = r$ and $\mathcal{P}'$ is a crosscap, or*

(ii) *$|\mathcal{P}'| = s$ and $\mathcal{P}'$ is a leap, or*

(iii) *$|\mathcal{P}'| = t$ and $\mathcal{P}'$ is a doublecross.*

Proof. Choose $Q_1, \ldots, Q_n \in \mathcal{P}$ in order with n maximum. By a result of [1], if $n(r-1) < |\mathcal{P}|$ then (i) is true. We assume then that $n \geq 2s+t-7$. From the maximality of n and the fact that $\mathcal{P}$ is crooked, there exists $P \in \mathcal{P}$

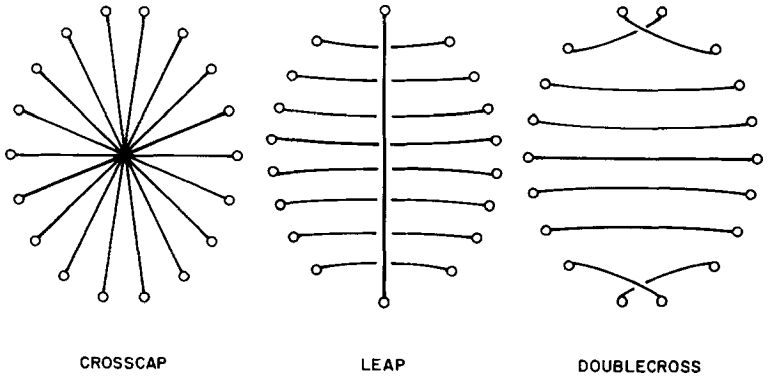


FIG. 3. Three crooked transactions.

[p. 60] crossing Q_1. If P crosses Q_{s-1} then (ii) is true. We assume then that it does not, and so P does not cross any Q_j for $s-1 \leq j \leq n$. Similarly, we may assume that there exists $P' \in \mathscr{P}$ crossing Q_n but not crossing any Q_j for $1 \leq j \leq n-s+2$. Now $t \geq 4$, and so $s-1 \leq n-s+2$. Hence neither P nor P' crosses Q_{s-1}, and so P does not cross P'. It follows that

$$\{P, Q_1, Q_{s-1}, Q_s, \ldots, Q_{s+t-7}, Q_{s+t-6}, Q_n, P'\}$$

is a doublecross and (iii) is true. □

8. Societies with No Large Transaction

The other possible outcome of (6.1), that (G, Ω) has sophistication at most $3p+7$, can also be made more concrete at some sacrifice of sharpness. To do so we investigate the structure of societies with no large transaction.

Let (G, Ω) be a society. A *linear decomposition* of (G, Ω) is an enumeration $\{t_1, \ldots, t_n\}$ of $\bar{\Omega}$ where $(t_1, \ldots, t_n)$ is clockwise, together with a family $(X_i : 1 \le i \le n)$ of subgraphs of G, with the following properties:

(i) $\bigcup(X_i : 1 \le i \le n) = G$

(ii) for $1 \le i < i' \le n$, $E(X_i \cap X_{i'}) = \emptyset$

(iii) for $1 \le i \le n$, $t_i \in V(X_i)$

(iv) for $1 \le i \le i' \le i'' \le n$, $X_i \cap X_{i''} \subseteq X_{i'}$.

The *depth* of such a linear decomposition is

$$\max(|V(X_i \cap X_{i'})| : 1 \le i < i' \le n).$$

It is easy to see that every society with a linear decomposition of depth $\le k$ has no transaction of cardinality $> 2k$. (We omit the proof because we shall not need the result.) What concerns us here is a partial converse, the following.

(8.1) *Every society (G, Ω) with $\bar{\Omega} \neq \emptyset$ or $V(G) = \emptyset$, and with no transaction of cardinality $> k$ has a linear decomposition of depth $\le k$.*[F2]

Proof. Let (G, Ω) be a society with no transaction of cardinality $> k$, and let $\bar{\Omega} = \{t_1, \ldots, t_n\}$, where $(t_1, \ldots, t_n)$ is clockwise. For $1 \le j \le n-1$, let $A_j = \{t_1, \ldots, t_j\}$, $B_j = \{t_{j+1}, \ldots, t_n\}$, so that (A_j, B_j) is a fence of Ω. Choose k_j maximum so that there are k_j mutually disjoint $A_j \to B_j$ paths in G. Let (Y_j, Z_j) be a separation of G satisfying the following conditions (1) and (2).

(1) $A_j \subseteq V(Y_j)$, $B_j \subseteq V(Z_j)$, and $|V(Y_j \cap Z_j)| = k_j$.

(This is possible by Menger's theorem.)

[p. 61] (2) *Subject to (1), Y_j is minimal.*

We claim that $Y_1 \subseteq Y_2 \subseteq \cdots \subseteq Y_{n-1}$ and $Z_1 \supseteq Z_2 \supseteq \cdots \supseteq Z_{n-1}$. For let i, j satisfy $1 \le i \le j \le n-1$. Then $(Y_i \cap Y_j, Z_i \cup Z_j)$ is a separation of G with $A_i \subseteq V(Y_i \cap Y_j)$, $B_i \subseteq V(Z_i \cup Z_j)$, and hence

$$|V((Y_i \cap Y_j) \cap (Z_i \cup Z_j))| \ge k_i.$$

Similarly

$$|V((Y_i \cup Y_j) \cap (Z_i \cap Z_j))| \ge k_j.$$

[F2] **Degenerate case.** As printed, (8.1) fails when $\bar{\Omega} = \emptyset$ and G is not null. Then every transaction is empty, so there is no transaction of cardinality $> k$. But a linear decomposition has $n = |\bar{\Omega}| = 0$, and then condition (i) asks that the empty union be G, which is false. With the added hypothesis (8.1) holds. The use of (8.1) in Graph Minors XV assumes $\bar{\Omega} \neq \emptyset$.

Thus

$$\begin{aligned} k_i + k_j &\le |V((Y_i \cap Y_j) \cap (Z_i \cup Z_j))| + |V((Y_i \cup Y_j) \cap (Z_i \cap Z_j))| \\ &\le |V(Y_i \cap Z_i)| + |V(Y_j \cap Z_j)| = k_i + k_j. \end{aligned}$$

Hence we have equality throughout, and in particular $|V((Y_i \cap Y_j) \cap (Z_i \cup Z_j))| = k_i$. Thus from condition (2), $Y_i \cap Y_j = Y_i$, and so $Y_i \subseteq Y_j$. Furthermore

$$\begin{aligned} k_i &= |V((Y_i \cap Y_j) \cap (Z_i \cup Z_j))| = |V(Y_i \cap (Z_i \cup Z_j))| \\ &= |V(Y_i \cap Z_i)| + |V(Y_i \cap Z_j) - V(Z_i)| = k_i + |V(Y_i \cap Z_j) - V(Z_i)|. \end{aligned}$$

Hence $V(Y_i \cap Z_j) \subseteq V(Z_i)$. But $V(Y_i) \cup V(Z_i) = V(G)$ and so $V(Z_j) \subseteq V(Z_i)$. Moreover, $E(Z_j) = E(G) - E(Y_j)$, $E(Z_i) = E(G) - E(Y_i)$, and $E(Y_i) \subseteq E(Y_j)$, and so $E(Z_j) \subseteq E(Z_i)$. Hence $Z_j \subseteq Z_i$. This proves our claim that $Y_1 \subseteq Y_2 \subseteq \cdots \subseteq Y_{n-1}$ and $Z_1 \supseteq Z_2 \supseteq \cdots \supseteq Z_{n-1}$.

Let $Y_n = Z_0 = G$, and let Y_0 and Z_n be the null graph. For $1 \le i \le n$, let $X_i = Y_i \cap Z_{i-1}$. We claim that $(X_i : 1 \le i \le n)$ is a linear decomposition of (G, Ω). We verify conditions (i), ..., (iv) in the definition of such a decomposition.

Condition (i). Let $v \in V(G)$, and choose i with $1 \le i \le n$, minimum such that $v \in V(Y_i)$. Then $v \notin V(Y_{i-1})$ and so $v \in V(Z_{i-1})$; hence $v \in V(X_i)$. Thus $V(X_1 \cup \cdots \cup X_n) = V(G)$. A similar argument shows that $E(X_1 \cup \cdots \cup X_n) = E(G)$ and so $X_1 \cup \cdots \cup X_n = G$.

Condition (ii). Suppose that $e \in E(X_i \cap X_j)$ where $1 \le i < j \le n$. Then $e \in E(Y_i)$ and $e \in E(Z_{j-1}) \subseteq E(Z_i)$, which contradicts the fact that $E(Y_i \cap Z_i) = \emptyset$.

Condition (iii). For i with $1 \le i \le n$, we have $t_i \in V(Y_i)$ and $t_i \in V(Z_{i-1})$. Hence $t_i \in V(X_i)$.

Condition (iv). For i, i', i'' with $1 \le i \le i' \le i'' \le n$, $X_i \cap X_{i''} \subseteq Y_i \subseteq Y_{i'}$ and $X_i \cap X_{i''} \subseteq Z_{i''-1} \subseteq Z_{i'-1}$ and so $X_i \cap X_{i''} \subseteq X_{i'}$, as required.

[p. 62] Hence $(X_i : 1 \le i \le n)$ is a linear decomposition of (G, Ω). We now examine its depth. Suppose that $1 \le i < j \le n$. Now $X_i \cap X_j \subseteq Y_i \cap Z_{j-1} \subseteq Y_i \cap Z_i$ and so

$$|V(X_i \cap X_j)| \le |V(Y_i \cap Z_i)| = k_i \le k.$$

Hence the decomposition has depth $\le k$. This completes the proof. □

9. Removing a Rural Neighborhood

The result of this section is the following.

(9.1) *If the composition of a society (G, Ω) with some rural neighborhood has a crooked transaction of cardinality $\geq p$, then so does (G, Ω).*

Proof. Let us choose a rural neighborhood (G_1, Ω_1, Ω) so that the following hold.

(1) *The composition of (G, Ω) with (G_1, Ω_1, Ω) has a crooked transaction $\mathscr{P}$ of cardinality $\geq p$.*

(2) *Subject to (1), (G_1, Ω_1, Ω) and $\mathscr{P}$ are chosen with $|V(G_1)| + |E(G_1)| + |E(\bigcup \mathscr{P})| - \frac{1}{2}|\bar{\Omega}_1|$ minimum.*

We observe

(3) *Every edge of G_1 is an edge of $\bigcup \mathscr{P}$, and every vertex of G_1 not in $V(G)$ is a vertex of $\bigcup \mathscr{P}$.*

For we could delete any other edge or vertex, contradicting (2).

(4) *For any $P \in \mathscr{P}$, at most one internal vertex of P is in $\bar{\Omega}_1$, and the subpaths of P from such a vertex to the ends of P both have an edge in $E(G)$.*

For let $c \in \bar{\Omega}_1$ be an internal vertex of P. Let Q be the path with $V(Q) = \{c\}$; then Q is an attachment to $\mathscr{P}$ with leg P. If it is exchangeable, there is a crooked transaction $\mathscr{P}'$ of cardinality $\geq p$ such that $\bigcup \mathscr{P}'$ has fewer edges than $\bigcup \mathscr{P}$, contrary to (2). Thus, by (3.2), P is an overpass; and since Q is not exchangeable, neither of the subpaths of P from c to the ends of P cross every other member of $\mathscr{P}$, and so both cross at least one. Hence neither of these paths is separating, and so neither is a subgraph of G_1, and so both have an edge in $E(G)$. Finally, if more than one internal vertex of P is in $\bar{\Omega}_1$, then by (3.3) there is a crooked transaction $\mathscr{P}'$ with $|\mathscr{P}'| \geq p + 1$ such that $\bigcup \mathscr{P}'$ has fewer edges than $\bigcup \mathscr{P}$, contrary to (2). This proves (4).

[p. 63] (5) *For any $P \in \mathscr{P}$, the first and last edges of P are in $E(G)$, and hence $V(\mathscr{P}) \subseteq \bar{\Omega}$.*

For let a be the first vertex of P, and b the second. Let e be the edge of P joining them, and suppose that $e \notin E(G)$. Now by (4), $b \notin \bar{\Omega}_1$, and so there is a cyclic permutation Ω_2 with

$$\bar{\Omega}_2 = (\bar{\Omega}_1 - \{a\}) \cup \{b\}$$

$$\Omega_2|(\bar{\Omega}_1 - \{a\}) = \Omega_1|(\bar{\Omega}_1 - \{a\})$$

and with (x, y, b) Ω_2-clockwise whenever (x, y, a) is Ω_1-clockwise. Then $(G_1 \setminus a, \Omega_2, \Omega)$ is a rural neighborhood, and the composition of (G, Ω) with it has a crooked transaction of cardinality $\geq p$, contrary to (2).

Now, to prove (9.1) take a plane Σ, and closed discs $\Delta_0 \subseteq \Delta \subseteq \Sigma$, and a drawing Γ of G_1 in Σ as in the definition of "rural neighborhood." Let I

be the infinite component of $\Sigma - (\Delta_0 \cup \Gamma)$. Then all vertices of G_1 which are drawn in $bd(I)$ belong to $\bar{\Omega}_1$, because otherwise we could enlarge $\bar{\Omega}_1$, contrary to (2). Suppose that $E(G_1) \neq \emptyset$. Then some edge e of G_1 must be drawn in $bd(I)$, and so its end-vertices must belong to $\bar{\Omega}_1$. By (3), e is an edge of some $P \in \mathscr{P}$ and by (5) it is not the first or last edge of P. Therefore both end-vertices of e are internal vertices of P which belong to $\bar{\Omega}_1$, contradicting (4). This contradiction shows that $E(G_1) = \emptyset$, and so $\mathscr{P}$ is a crooked transaction in (G, Ω), as required. □

10. Exploiting a Rural Neighborhood

Let (G, Ω, Ω_0) be a rural neighborhood. Then there is a drawing Γ in a plane Σ and discs Δ, Δ_0 as in the definition of "rural" in Section 1. We call $(\Sigma, \Gamma, \Delta, \Delta_0)$ a *presentation* of (G, Ω, Ω_0). For a fixed presentation and integer $s \geq 0$, an *s-nest* is a sequence $(C_1, \ldots, C_s)$ of circuits of G, mutually disjoint, such that $\Delta_0 \subseteq \Delta_1 \subseteq \cdots \subseteq \Delta_s \subseteq \Delta$, where Δ_i denotes the closed disc in Σ bounded by the image under Γ of C_i $(1 \leq i \leq s)$. If there is an s-nest in some presentation of (G, Ω, Ω_0), we say that (G, Ω, Ω_0) has *breadth* $\geq s$.

Again, let (G, Ω, Ω_0) be a neighborhood. We say that $X \subseteq \bar{\Omega}$ is *weakly-linked* if there are $|X|$ mutually disjoint $X \to \bar{\Omega}_0$ paths of G, and *linked* if these paths can be chosen such that each has only its initial vertex in $\bar{\Omega}$. The object of this section is, roughly, to prove that if there is a crooked transaction $\mathscr{P}$ of cardinality p in the composition of some society with some rural neighborhood (G, Ω, Ω_0) of sufficient breadth, and $X \subseteq \bar{\Omega}$ is linked and $|X|$ is sufficiently large, then $\mathscr{P}$ may be chosen with $V(\mathscr{P}) \subseteq X$. We begin with the following lemma.

[p. 64] (10.1) *Let (G, Ω, Ω_0) be a rural neighborhood and let $(\Sigma, \Gamma, \Delta, \Delta_0)$ be a presentation with an s-nest. Let $X \subseteq \bar{\Omega}$ be weakly-linked, with $|X| = r$. Then we may choose an s-nest $(C_1, \ldots, C_s)$ and disjoint paths $P_1, \ldots, P_r$ of G each with one end in X and the other in $\bar{\Omega}_0$, such that for $1 \leq i \leq r$ and $1 \leq j \leq s$, $P_i \cap C_j$ is a path.*

The proof of (10.1) is very similar to the proof of [2, Theorem (4.1)]~~, and we omit it~~, and we give it now.

We give the proof, following the proof of [2, Theorem (4.1)]. Since X is weakly-linked there are r mutually disjoint $X \to \bar{\Omega}_0$ paths, and by hypothesis there is an s-nest. Choose an s-nest $(C_1, \ldots, C_s)$ and mutually disjoint $X \to \bar{\Omega}_0$ paths $P_1, \ldots, P_r$ such that $|E(C_1 \cup \cdots \cup C_s \cup P_1 \cup \cdots \cup P_r)|$ is minimum. For $1 \leq j \leq s$ let Δ_j be the closed disc bounded by the image of C_j, as in the definition of an s-nest. Suppose that $P_i \cap C_j$ is not a path

for some i, j, and choose such a j as large as possible. Each P_i meets C_j, since it runs from a vertex in $bd(\Delta)$, which is not in the interior of Δ_j, to a vertex in $\Delta_0 \subseteq \Delta_j$. We call a subpath P of some $P_{i'}$ a *detour* if its ends are distinct vertices of C_j and no other vertex and no edge of P is in C_j. A detour is drawn either in Δ_j, or outside Δ_j except for its ends.

(1) *No detour is drawn outside Δ_j except for its ends.*

For suppose that a detour $P \subseteq P_{i'}$ with ends v, v' is drawn in this way. First, P is disjoint from C_m for $m \neq j$. For $m < j$ this holds since C_m is drawn in the interior of Δ_j and $v, v' \notin V(C_m)$. Let $m > j$, and suppose that P has a vertex w in C_m. The path $P_{i'}$ starts outside the interior of Δ_m, and so it meets C_m before it reaches v and v', which are drawn in the interior of Δ_m. By the choice of j, $P_{i'} \cap C_m$ is a path, and so it contains the part of $P_{i'}$ between this first vertex and w, which contains v or v'. This is impossible, since $v, v' \notin V(C_m)$. Now let Q_1, Q_2 be the two paths of C_j between v and v'. Then $P \cup Q_1$ and $P \cup Q_2$ are circuits, and since P is drawn outside Δ_j, one of them, say $P \cup Q_1$, bounds a closed disc that includes Δ_j. This disc is included in Δ_{j+1} if $j < s$, since P misses C_{j+1}, and in Δ if $j = s$. So replacing C_j by $P \cup Q_1$ gives an s-nest. Some edge of Q_2 is in none of $P_1, \ldots, P_r$. For otherwise, since these paths are disjoint and Q_2 is connected, Q_2 would be a subpath of $P_{i'}$ between v and v' other than P, which is impossible. Hence the replacement makes the union smaller, contrary to the choice of the nest and the paths. This proves (1).

(2) *For $1 \leq i' \leq r$ and each vertex u of $P_{i'} \cap C_j$, the subpath of $P_{i'}$ from u to its end in $\bar{\Omega}_0$ is drawn in Δ_j.*

For if it leaves Δ_j then it comes back, since its end is in $\Delta_0 \subseteq \Delta_j$, and so it contains a detour drawn outside Δ_j except for its ends, contrary to (1).

Now $P_i \cap C_j$ is non-null and not a path, and so some subpath of P_i between two vertices of C_j is not included in C_j. A minimal such subpath P is a detour, with ends v, v' say, and by (1) it is drawn in Δ_j. Let Q_1, Q_2 be the two paths of C_j between v and v'. Then P splits Δ_j into two closed discs, bounded by $P \cup Q_1$ and $P \cup Q_2$. The drawing uses no point in the interior of Δ_0, and so we may choose $k \in \{1, 2\}$ such that every connected subset of Δ_j that misses P and meets Q_k is disjoint from Δ_0. Then no $P_{i'}$ with $i' \neq i$ meets Q_k. For if u is a common vertex, then by (2) the subpath of $P_{i'}$ from u to its end in $\bar{\Omega}_0$ is drawn in Δ_j, and it misses P, since $P_{i'}$ and P_i are disjoint, and this contradicts the choice of k. Now replace P by Q_k in P_i. The result contains an $X \to \bar{\Omega}_0$ path P_i', with edges in $(E(P_i) - E(P)) \cup E(Q_k)$, and P_i' is disjoint from the other paths $P_{i'}$. The edge of P incident with v is in no circuit C_m, since it is not in C_j and the other circuits do not contain v, and it is in no $P_{i'}$ with $i' \neq i$. Hence replacing P_i by P_i' makes the union

smaller, again contrary to the choice. This proves (10.1).[F3]

We deduce

(10.2) *Let (G, Ω, Ω_0) be a rural neighborhood of breadth $\geq s$, and let $\{a_1, a_2, \ldots, a_s, b_1, b_2, \ldots, b_s\} \subseteq \bar{\Omega}$ be weakly-linked, where $(a_s, a_{s-1}, \ldots, a_1, b_1, b_2, \ldots, b_s)$ is clockwise. Then there is a transaction $\{Q_1, \ldots, Q_s\}$ in (G, Ω) such that Q_i has ends a_i, b_i and Q_i meets every path from $\Omega(a_i, b_i)$ to $\Omega(b_i, a_i) \cup \bar{\Omega}_0$ $(1 \leq i \leq s)$.*

Proof. We take a presentation $(\Sigma, \Gamma, \Delta, \Delta_0)$ with an s-nest, put $r = 2s$ and $X = \{a_1, \ldots, a_s, b_1, \ldots, b_s\}$, and choose $C_1, \ldots, C_s$ and $P_1, \ldots, P_r$ as in (10.1). We number $P_1, \ldots, P_r$ so that P_i has initial vertex a_i $(1 \leq i \leq s)$ and b_{i-s} $(s+1 \leq i \leq r)$. For $1 \leq i \leq s$, let Q_i be the union of the minimal subpath of P_i from a_i to $V(C_{s+1-i})$, the minimal subpath of P_{s+i} from b_i to $V(C_{s+1-i})$, and the subpath of C_{s+1-i} from the end of the first path to the end of the second, moving clockwise around C_{s+1-i}, that is, the one for which the disc bounded by C_{s+1-i} lies on the same side of Q_i as $\Omega(b_i, a_i)$.[F4] Then $\{Q_1, \ldots, Q_s\}$ is the desired transaction. □

We come now to a preliminary form of the main result of this section.

(10.3) *Let $p \geq 2$ with $p \neq 3$. Let (G, Ω, Ω_0) be a rural neighborhood of breadth $\geq p + 4$, and let $X \subseteq \bar{\Omega}$ be weakly-linked, with $|X| \geq 2p + 8$. Let $(G \cup G_0, \Omega)$ be the composition of some society (G_0, Ω_0) with (G, Ω, Ω_0), and suppose that there is a crooked transaction $\mathscr{P}$ in $(G \cup G_0, \Omega)$ with $|\mathscr{P}| \geq p$. Then $\mathscr{P}$ may be chosen with $V(\mathscr{P}) \subseteq X$.*

Proof. Let $X \subseteq Y \subseteq \bar{\Omega}$, and let $\mathscr{P}$ be a crooked transaction in $(G \cup G_0, \Omega)$ with $|\mathscr{P}| \geq p$ and $V(\mathscr{P}) \subseteq Y$. We show that $\mathscr{P}$ may be chosen with $V(\mathscr{P}) \subseteq X$. The theorem is the case $Y = \bar{\Omega}$. We proceed by induction on $|\bar{\Omega} Y - X|$. If this number is zero the result is trivial, and so we assume that there exists $x \in \bar{\Omega} Y - X$. ~~Let $\Omega' = \Omega \restriction (\bar{\Omega} - \{x\})$. Then (G, Ω', Ω_0) is a rural neighborhood of breadth $\geq p + 4$, and $X \subseteq \bar{\Omega}'$. But $|\bar{\Omega}' - X| < |\bar{\Omega} - X|$, and so from our inductive hypothesis it suffices to show that $(G \cup G_0, \Omega')$ has a crooked transaction of cardinality $\geq p$, that is,~~By our inductive hypothesis, applied to $Y - \{x\}$, it suffices to show that $(G \cup G_0, \Omega)$ has a crooked transaction $\mathscr{P}$ of cardinality $\geq p$ with ~~$x \notin V(\mathscr{P})$~~$V(\mathscr{P}) \subseteq Y - \{x\}$.[F5] Put

[F3] **Missing proof.** The paper omits the proof of (10.1). The proof above is the one used in the formalization.

[F4] **Proof gap.** As printed, Q_i runs along C_i, where C_1 is the innermost circuit of the nest. Then for $s \geq 2$ the path Q_1 runs from a_1 through C_s to C_1, while Q_s runs along C_s from P_s to P_{2s}, past the vertices where P_1 meets C_s, so Q_1 and Q_s meet. With C_{s+1-i} the path with the shortest interval $\Omega(a_i, b_i)$ runs along the outermost circuit, and the paths are disjoint.

[F5] **Proof gap.** As printed, the proof passes to (G, Ω', Ω_0) and calls it a rural neighborhood.

$s = p + 4$, and choose $2s$ elements $a_1, \ldots, a_s, b_1, \ldots, b_s$ of X so that

$$(a_s, a_{s-1}, \ldots, a_1, x, b_1, b_2, \ldots, b_s)$$

is clockwise. By (10.2) there is a transaction $\{Q_1, \ldots, Q_s\}$ in (G, Ω) such that for $1 \le i \le s$, Q_i ends a_i, b_i and Q_i meets every path in G from $\Omega(a_i, b_i)$ to $\Omega(b_i, a_i) \cup \bar{\Omega}_0$.

[p. 65] (1) *If P is an Ω-path in $G \cup G_0$ and $P \cap Q_i$ is null for some i $(1 \le i \le s)$ and some end of P is in $\Omega(a_i, b_i)$ then both ends of P are in $\Omega(a_i, b_i)$ and no vertex of P is in G_0.*

For otherwise P has a vertex in $\Omega(a_i, b_i)$ and a vertex in $\Omega(b_i, a_i) \cup V(G_0)$. Let us choose a minimal subpath P' of P with the same property. Then at most one vertex of P' is in $V(G_0)$, and so every edge of P' is in $E(G)$; thus $P' \subseteq G$ and P' has a vertex in $\Omega(b_i, a_i) \cup \bar{\Omega}_0$. But Q_i meets every path of G from $\Omega(a_i, b_i)$ to $\Omega(b_i, a_i) \cup \bar{\Omega}_0$, and Q_i does not meet P' since $P' \subseteq P$, a contradiction.

(2) *If $\{P, P'\}$ is a cross in $(G \cup G_0, \Omega)$ and $P \cap Q_i, P' \cap Q_i$ are both null for some i $(1 \le i \le s)$ then no end of P or P' is in $\Omega(a_i, b_i)$.*

For suppose some end a of P is in $\Omega(a_i, b_i)$. By (1), the other end b of P is in $\Omega(a_i, b_i)$, and since $\{P, P'\}$ is a cross there is an end of P' in $\Omega(a_i, b_i)$ between a and b. By (1), $P, P' \subseteq G$ and so $\{P, P'\}$ is a cross of (G, Ω), a contradiction since (G, Ω, Ω_0) (and hence (G, Ω)) is rural.

(3) *There is a crooked transaction $\mathscr{P}$ in $(G \cup G_0, \Omega)$ of cardinality p with ~~$x \notin V(\mathscr{P})$~~ $V(\mathscr{P}) \subseteq Y - \{x\}$.*

For there is a separation (G_1, G_2) of G with $V(G_1 \cap G_2) = V(Q_s)$, $\Omega(a_s, b_s) \subseteq V(G_1)$, $\Omega(b_s, a_s) \subseteq V(G_2)$ and $\bar{\Omega}_0 \subseteq V(G_2)$, by our choice of Q_s. Then $(G_1, G_2 \cup G_0)$ is a separation of $G \cup G_0$, and $V(G_1 \cap (G_2 \cup G_0)) = V(Q_s)$ since $V(G_1 \cap G_0) \subseteq \bar{\Omega}_0 \subseteq V(G_2)$. By (2), there is no cross $\{P, P'\}$ of $(G \cup G_0, \Omega)$ with $P \cap (G_2 \cup G_0), P' \cap (G_2 \cup G_0)$ both null. But $x \in \Omega(a_1, b_1)$ and so (3) follows from (5.2), applied to the society $(G \cup G_0, \Omega \mid Y)$. Its crosses and crooked transactions are the crosses and crooked transactions of $(G \cup G_0, \Omega)$ with all ends in Y, since $\Omega \mid Y$ orders Y as Ω does. So $\mathscr{P}$ is a crooked transaction in it, $\{Q_1, \ldots, Q_s\}$ is a transaction in it since $a_i, b_i \in X \subseteq Y$, and $x \in Y$. A crooked transaction given by (5.2)(ii) has no end in $\Omega(a_1, b_1)$, and so all its ends are in $Y - \{x\}$. We use (5.2) in the following form. The paths Q_i come from (10.2), and they may have internal vertices in $\bar{\Omega}$, which (5.2) does not allow. But (5.2) remains true without

But x is drawn in $bd(\Delta)$ and is not in $\bar{\Omega}'$, so the presentation of (G, Ω, Ω_0) is not a presentation of (G, Ω', Ω_0), and a new one is needed, which the proof does not give. Here the neighborhood (G, Ω, Ω_0) is kept, and only the set Y that holds the ends shrinks.

the hypothesis that $Q_1, \ldots, Q_{p+4}$ have no internal vertex in $\bar{\Omega}$. To see this, let (G', Ω') be the society obtained from (G, Ω) by adding, for each $v \in \bar{\Omega}$, a new vertex v' and a new edge joining v and v', where $\bar{\Omega}'$ is the set of new vertices and $\Omega'(v') = u'$ when $\Omega(v) = u$. For an Ω-path P with ends a, b, let P' be the Ω'-path with ends a', b' obtained from P by adding the new edges at a and b. Since the new vertices have degree one, every Ω'-path is P' for exactly one Ω-path P, and no Ω'-path has an internal vertex in $\bar{\Omega}'$. This correspondence keeps crossing, and so it maps separating paths to separating paths, transactions to transactions, and crooked transactions to crooked transactions of the same cardinality, and $\Omega(a, b) \cap V(\mathcal{P})$ is empty if and only if $\Omega'(a', b') \cap V(\mathcal{P}')$ is empty. Given (G_1, G_2) as in (5.2), let G_1' be G_1 with the new edges at the vertices in $\Omega(a_{p+4}, b_{p+4}) \cup \{a_{p+4}, b_{p+4}\}$, and let G_2' be G_2 with the new edges at the vertices in $\Omega(b_{p+4}, a_{p+4})$ and with the vertices a'_{p+4}, b'_{p+4}. Then (G_1', G_2') is a separation of G' with $V(G_1' \cap G_2') = V(Q'_{p+4})$, $\Omega'(a'_{p+4}, b'_{p+4}) \subseteq V(G_1')$ and $\Omega'(b'_{p+4}, a'_{p+4}) \subseteq V(G_2')$. So (5.2) applies to (G', Ω') and $Q'_1, \ldots, Q'_{p+4}$, and its outcome (i) or (ii) there gives outcome (i) or (ii) for (G, Ω), since $P_1' \cap G_2'$ is null exactly when $P_1 \cap G_2$ is null[F6].

This completes the proof. □

The main result of this section is the following.

(10.4) *Let $p \geq 2$ with $p \neq 3$. Let (G, Ω, Ω_0) be a rural neighborhood of breadth $\geq p+5$, and let $X \subseteq \bar{\Omega}$ be linked with $|X| \geq 2p+8$. Let $(G \cup G_0, \Omega)$ be the composition of some society (G_0, Ω_0) with (G, Ω, Ω_0), and suppose that there is a crooked transaction $\mathscr{P}$ in $(G \cup G_0, \Omega)$ with $|\mathscr{P}| \geq p$. Then $\mathscr{P}$ may be chosen with $V(\mathscr{P}) \subseteq X$ and such that no member of $\mathscr{P}$ has any internal vertex in $\bar{\Omega}$.*

Proof. Take a presentation $(\Sigma, \Gamma, \Delta, \Delta_0)$ of (G, Ω, Ω_0) with a $(p+5)$-nest $(C_1', \ldots, C'_{p+5})$. Then $(C_1', \ldots, C'_{p+4})$ is a $(p+4)$-nest and $V(C_1'), \ldots, V(C'_{p+4})$ are all disjoint from $\bar{\Omega}$. Since X is linked, there are mutually disjoint $X \to \bar{\Omega}_0$ paths $P_1', \ldots, P'_{2p+8}$ of G each with no internal vertex in $\bar{\Omega}$. Let G' be obtained from G by deleting every edge of G with an end in $\bar{\Omega}$ except for the first edges of $P_1', \ldots, P'_{2p+8}$. Then $C_1', \ldots, C'_{p+4}$ and $P_1', \ldots, P'_{2p+8}$ are all subgraphs of G'. By (10.1), applied to (G', Ω, Ω_0), [p. 66] there is a $(p+4)$-nest $(C_1, \ldots, C_{p+4})$ and disjoint $X \to \bar{\Omega}_0$ paths $P_1, \ldots, P_{2p+8}$ of G', such that for $1 \leq i \leq 2p+8$ and $1 \leq j \leq p+4$, $P_i \cap C_j$ is a path. Since every vertex of G' in $\bar{\Omega}$ has valency ≤ 1 it follows that for $1 \leq j \leq p+4$ no vertex of C_j is in $\bar{\Omega}$ and that for $1 \leq i \leq 2p+8$ no vertex of P_i is in $\bar{\Omega}$

[F6] **Proof gap.** As printed, (5.2) does not apply, since the paths of (10.2) may have internal vertices in $\bar{\Omega}$. The form of (5.2) given here, without that hypothesis, is the one proved and used in the formalization.

except its initial vertex. Let x_i be the initial vertex of P_i and let y_i be the first vertex of P_i in $V(C_{p+4})$ $(1 \le i \le 2p+8)$. Put $Y = \{y_1, \ldots, y_{2p+8}\}$. Let G_1 be the subgraph of G drawn in the closed disc Δ_1 bounded by the drawing of C_{p+4} (thus, $C_{p+4} \subseteq G_1$). Let Ω_1 be the cyclic permutation in $V(G_1)$ with $\bar{\Omega}_1 = V(C_{p+4})$ corresponding to the clockwise orientation of $bd(\Delta_1)$. Then $(G \cup G_0, \Omega)$ is the composition of $(G_1 \cup G_0, \Omega_1)$ with some rural neighborhood, and so by (9.1), $(G_1 \cup G_0, \Omega_1)$ has a crooked transaction of cardinality $\ge p$. But $Y \subseteq \bar{\Omega}_1$ is weakly-linked in the rural neighborhood $(G_1, \Omega_1, \Omega_0)$, because our choices of $C_1, \ldots, C_{p+4}$ and $P_1, \ldots, P_{2p+8}$ ensure that for $1 \le i \le 2p+8$ the subpath of P_i from Y to $\bar{\Omega}_0$ is a path of G_1. This neighborhood has breadth $\ge p+4$ and so by (10.3), there is a crooked transaction $\mathscr{P}$ in $(G_1 \cup G_0, \Omega_1)$ of cardinality $\ge p$ with $V(\mathscr{P}) \subseteq Y$. For each $P \in \mathscr{P}$, let its ends be $y_i, y_{i'} \in Y$, where $1 \le i, i' \le 2p+8$, and let P^+ denote the union of P with the subpath of P_i between x_i and y_i and the subpath of $P_{i'}$ between $x_{i'}$ and $y_{i'}$. Then P^+ is a path of $G \cup G_0$[F7] with both ends in X and no internal vertex in $\bar{\Omega}$, and the set $\{P^+ : P \in \mathscr{P}\}$ is the desired crooked transaction. We check this. The subpath of P_i between x_i and y_i meets Δ_1 only in y_i, since y_i is the first vertex of P_i in $V(C_{p+4})$ and $x_i \in bd(\Delta)$ is not in Δ_1. So this subpath meets $G_1 \cup G_0$ only in y_i, because G_1 is drawn in Δ_1 and $V(G \cap G_0) = \bar{\Omega}_0$ is drawn in $\Delta_0 \subseteq \Delta_1$. Since also the members of $\mathscr{P}$ are disjoint and $P_1, \ldots, P_{2p+8}$ are disjoint, the paths P^+ $(P \in \mathscr{P})$ are mutually disjoint. Let H_2 be the subgraph of G drawn in the closure of $\Delta - \Delta_1$. Then (H_2, Ω, Ω_1) is a rural neighborhood, and the subpaths of $P_1, \ldots, P_{2p+8}$ between x_i and y_i are disjoint paths of H_2 from $\bar{\Omega}$ to $\bar{\Omega}_1$. Hence for distinct i_1, i_2, i_3, i_4, $\{x_{i_1}, x_{i_3}\}$ crosses $\{x_{i_2}, x_{i_4}\}$ in Ω if and only if $\{y_{i_1}, y_{i_3}\}$ crosses $\{y_{i_2}, y_{i_4}\}$ in Ω_1. A cyclic order of a finite set is determined, up to reversal, by which of its pairs cross. So the ends of the paths P^+ lie around Ω in the same cyclic order, up to reversal, as the corresponding ends of the members of $\mathscr{P}$ around Ω_1. Whether a set of disjoint paths is a transaction, and which of its members are peripheral, depends only on this order. Hence $\{P^+ : P \in \mathscr{P}\}$ is a crooked transaction in $(G \cup G_0, \Omega)$, of cardinality $|\mathscr{P}| \ge p$.[F8] This completes the proof. □

In later papers we shall also require the following, which we include here because of its similarity to (10.4).

(10.5) *Let $p \ge 0$ and let (G, Ω, Ω_0) be a rural neighborhood of breadth*

[F7] **Misprint.** Here P is a path of $G_1 \cup G_0$ and may use edges of G_0, so P^+ is a path of $G \cup G_0$, as a member of a transaction in $(G \cup G_0, \Omega)$ should be.

[F8] **Missing proof.** The paper says that the new paths form a crooked transaction and gives no reason. The reason above is the one used in the formalization.

$\geq \frac{1}{2}p + 1$. *Let* $X \subseteq \bar{\Omega}$ *be linked, with* $|X| \geq p$. *Let* $(G \cup G_0, \Omega)$ *be the composition of some society* (G_0, Ω_0) *with* (G, Ω, Ω_0), *and let* $Y \subseteq V(G_0)$. *Then either there are* p *mutually disjoint paths of* $G \cup G_0$ *from* X *to* Y, *each with only its initial vertex in* $\bar{\Omega}$, *or there exists* $W \subseteq V(G \cup G_0)$ *with* $|W| < p$ *and* $W \cap \bar{\Omega} = \emptyset$ *such that every path of* $G \cup G_0$ *from* $\bar{\Omega}$ *to* Y *has a vertex in* W.

Proof. Suppose that there is no W as in the theorem. Let $G' = (G \cup G_0) \setminus (\bar{\Omega} - X)$. It suffices to show that there are p mutually disjoint $X \to Y$ paths in G'; and therefore, by Menger's theorem, it suffices to show that if $Z \subseteq V(G')$ and $|Z| < p$ then there is a path of G' from X to Y avoiding Z. Thus, let $Z \subseteq V(G')$ with $|Z| < p$. Let $s = \lceil \frac{1}{2}p \rceil$; as in (10.4), there is an s-nest $(C_1, \ldots, C_s)$ in some presentation of (G, Ω, Ω_0), such that $C_1, \ldots, C_s \subseteq G'$ and $V(C_s) \cap \bar{\Omega} = \emptyset$. Since X is linked and $|X| \geq p$, there is a path P_1 of G' from X to $\bar{\Omega}_0$ which avoids Z. Since $C_1, \ldots, C_s$ are mutually disjoint, there exists j with $1 \leq j \leq s$ such that $|Z \cap V(C_j)| \leq 1$. Since there is no W as in the theorem, there are p paths of $G \cup G_0$ from $\bar{\Omega}$ to Y, mutually vertex-disjoint except possibly for their initial vertices, by a form of Menger's theorem. Since these all meet $V(C_j)$, there is a path P_2 of G' [p. 67] from $V(C_j)$ to Y which avoids Z. Since $|Z \cap V(C_j)| \leq 1$, there is a path P_3 of C_j from a vertex of P_1 to a vertex of P_2 which avoids Z. Then $P_1 \cup P_2 \cup P_3 \subseteq G'$ is connected and has a vertex in X and a vertex in Y. Thus there is an $X \to Y$ path $P \subseteq P_1 \cup P_2 \cup P_3$; and P avoids Z since P_1, P_2, P_3 each avoid Z. This completes the proof. □

11. Coping with Connectivity Failure

The object of this section is to prove a version of our main theorem (6.1) which does not require the hypothesis that (G, Ω) be 3-connected and internally 4-connected. To achieve this we shall need several lemmas.

Let $(\Sigma, \Gamma, \Delta, \Delta_0)$ be a presentation of a rural neighborhood (G, Ω, Ω_0), where G is simple. For any path P of G with distinct ends in $\bar{\Omega}_0$ and with no internal vertex in $\bar{\Omega}_0$ there is a unique closed disc $\Delta(P) \subseteq \Delta$ bounded by the image of P under Γ and a portion of $bd(\Delta_0)$ such that $\Delta_0 \not\subseteq \Delta(P)$. A *prominence* is a path P of G with the following properties:

(i) the ends of P are distinct and are both in $\bar{\Omega}_0$

(ii) no internal vertex of P is in $\bar{\Omega}_0$

(iii) P has at most two edges

(iv) subject to (i), (ii), and (iii), $\Delta(P)$ is maximal.

If P is a prominence, $\Delta(P)$ is a *lawn*.

(11.1) *If L, L' are distinct lawns then every element of $L \cap L'$ is the image under Γ of a common end of the two corresponding prominences.*

The proof is clear, although tedious to write out~~, and we omit it~~, and we give it now.

The definition of the discs $\Delta(P)$ presupposes that the drawing meets $bd(\Delta_0)$ only in the vertices of $\bar{\Omega}_0$, and we assume this. Then each path P satisfying (i), (ii) and (iii) is drawn as an arc with both ends in $bd(\Delta_0)$ and no other point in Δ_0, and we call it an *ear*. The set $\Delta_0 \cup \Delta(P)$ is a closed disc, whose boundary consists of the drawing of P and the portion of $bd(\Delta_0)$ not in $\Delta(P)$, and the interior of $\Delta(P)$ misses Δ_0. We use three facts about ears, and we prove them first. We use several times that a connected set that misses the boundary of a closed disc lies in its interior or outside it.

(a) *If the drawing of an ear P', except its ends, lies in $\Delta(P)$ for an ear P, then $\Delta(P') \subseteq \Delta(P)$.*

For the boundary of the closed disc $\Delta_0 \cup \Delta(P')$ lies in $\Delta_0 \cup \Delta(P)$, and so, in the plane, $\Delta_0 \cup \Delta(P') \subseteq \Delta_0 \cup \Delta(P)$. Since the interior of $\Delta(P')$ misses Δ_0, it lies in $\Delta(P)$, and so does its closure $\Delta(P')$.

(b) *If the drawings of two ears P, P' meet only in common ends, then one of $\Delta(P), \Delta(P')$ includes the other, or they meet only in images of common ends of P and P'.*

Suppose that neither disc includes the other. The drawing of P' without its ends is connected and misses the boundary of $\Delta_0 \cup \Delta(P)$. If it lay in the interior, it would lie in $\Delta(P)$, since it misses Δ_0, and then $\Delta(P') \subseteq \Delta(P)$ by (a). So it lies outside $\Delta_0 \cup \Delta(P)$. Exchanging P and P', the drawing of P without its ends lies outside $\Delta_0 \cup \Delta(P')$. The interior of $\Delta(P)$ is connected and misses the boundary of $\Delta_0 \cup \Delta(P')$. It does not lie in $\Delta(P')$, since otherwise $\Delta(P)$, and so the drawing of P, would lie in $\Delta(P')$. Hence the interior of $\Delta(P)$ misses $\Delta(P')$, and exchanging P and P', the interior of $\Delta(P')$ misses $\Delta(P)$. Now let $z \in \Delta(P) \cap \Delta(P')$. By what we have shown, z is not in the drawing of P or of P' except as an end, and so $z \in bd(\Delta_0)$. Suppose that z is not the image of an end of P'. Then z is in the interior of the closed disc $\Delta_0 \cup \Delta(P')$, and so the points near z outside Δ_0 lie in $\Delta(P')$. But z is a limit of points of the interior of $\Delta(P)$, which are outside Δ_0 and not in $\Delta(P')$, a contradiction. So z is the image of an end of P', and in the same way of an end of P.

(c) *Let three arcs join a point $m \notin \Delta_0$ to distinct points x, y, z of $bd(\Delta_0)$, meeting each other only in m, and meeting Δ_0 only in x, y, z respectively. Then either the arc to z lies in the disc cut off by the arcs to x and y, or this disc is properly included in the disc cut off by the arcs to x and z, or in the disc cut off by the arcs to y and z.*

Here the disc cut off by two of the arcs is the closed disc bounded by their union and a portion of $bd(\Delta_0)$ that does not include Δ_0, as for ears. Let D be the disc cut off by the arcs to x and y, and let B be the portion of $bd(\Delta_0)$ between x and y not in D. If $z \notin B$, then the arc to z without its ends is connected and misses the boundary of the closed disc $\Delta_0 \cup D$, and near z it is in D, so it lies in D. Now let $z \in B$. Then the arc to z without its ends lies outside $\Delta_0 \cup D$, by the same fact, since near z it is outside $\Delta_0 \cup D$, and it splits the closure of the complement of $\Delta_0 \cup D$, in the plane with a point at infinity added, into two closed discs. One of them, D_1, is bounded by the arcs to x and z and the portion of B between x and z, and the other by the arcs to y and z and the rest of B. Suppose that D_1 does not contain the point at infinity. Then $D \cup D_1$ is a closed disc in the plane bounded by the arcs to y and z and a portion of $bd(\Delta_0)$, and it does not include Δ_0. So it is the disc cut off by the arcs to y and z, and it properly includes D, since D_1 has points outside D. In the other case, the same argument with x and y exchanged shows that D is properly included in the disc cut off by the arcs to x and z.

Now let P, P' be prominences with distinct lawns $L = \Delta(P)$ and $L' = \Delta(P')$. First we show that P and P' have no common middle vertex. Suppose that m is the middle vertex of both, and let x, y be the ends of P. We claim that every edge g of G joining m to a vertex $z \in \bar{\Omega}_0$ is drawn in L. If $z \in \{x, y\}$ then g is an edge of P, since G is simple. Otherwise the two paths formed by g and the edge of P at x, and by g and the edge of P at y, satisfy (i), (ii) and (iii). By (c), either g is drawn in L, or L is properly included in $\Delta(Q)$ for one of these two paths Q, which is impossible by (iv) for P. This proves the claim. Both edges of P' join m to vertices in $\bar{\Omega}_0$, and so by the claim P' is drawn in L. By (a), $L' \subseteq L$, and by (iv) for P', $L' = L$, a contradiction.

So P and P' have no common middle vertex, and $P \neq P'$ since $L \neq L'$. Hence their drawings meet only in common ends. For a common vertex that is not an end of one of them would be its middle vertex, and so it would not be in $\bar{\Omega}_0$, and it would be the middle vertex of the other too. And a common edge has an end that is the middle vertex of one of them, unless both P and P' consist of this edge alone. Now by (b), one of L, L' includes the other, or they meet only in images of common ends of P and P'. In the first case, if say $L \subseteq L'$, then $L = L'$ by (iv) for P, a contradiction. So every element of $L \cap L'$ is the image of a common end of P and P'.[F9]

[F9] **Missing proof.** The paper omits the proof of (11.1). The argument above is the one used in the formalization.

It follows that the union of Δ_0 with all lawns is a closed disc; we call its boundary the *wall*. Let $e \in E(G)$; we call it an *internal*, *external*, or *wall edge* if the arc which represents e in the drawing Γ is a subset of the interior of the disc bounded by the wall, the complement in Δ of this disc, or the wall itself, respectively. Every edge of G is of one of these three types.

(11.2) *If e is an external edge of G with both ends drawn in the wall, then its ends are both middle vertices of prominences.*

Proof. Let P be the path formed by e. If $\Delta(P)$ were defined, it would be a subset of a lawn, which is false since e is external. Thus not both ends of P are in $\bar{\Omega}_0$, and we may choose a prominence P_1 such that its middle vertex v_1 is an end of e. Let v_2 be the other end of e. Since G is simple and e is not a wall edge, v_2 is not an end of P_1. Let e_1 be an edge of P_1, and [p. 68] let P' be the path formed by e, e_1. Again, since $\Delta(P')$ is not a subset of a lawn it is not defined, and so $v_2 \notin \bar{\Omega}_0$. The result follows. □

A triangle in G is a circuit of length 3.

(11.3) *No triangle in G has an internal edge and an external edge.*

Proof. Let T be a triangle of G with $V(T) = \{v_1, v_2, v_3\}$, $E(T) = \{e_1, e_2, e_3\}$ where e_i is incident with v_j for $1 \le i, j \le 3$ if $i \ne j$. Let e_3 be an external edge and e_2 an internal edge. If both ends v_1, v_2 of e_3 are drawn in the wall, then by (11.2) there are prominences P_1, P_2 such that v_i is the middle vertex of P_i $(i = 1, 2)$. Since e_2 is internal, it is drawn in $\Delta(P_1)$, and so v_3 is drawn in $\Delta(P_1)$. But e_1 is not drawn in $\Delta(P_1)$ since v_2 is not drawn in $\Delta(P_1)$, and so v_3 is drawn in $bd(\Delta(P_1))$, and indeed $v_3 \in V(P_1)$. It follows that there is an edge of P_1 joining v_1 and v_3, a contradiction since G is simple and e_2 is internal.

Thus not both ends of e_3 are drawn in the wall. Certainly v_1 is drawn in the wall since e_2 is internal, and so v_2 is not drawn in the wall. Therefore e_1 as well as e_3 is external. Since e_2 is internal, both v_1 and v_3 are drawn in the wall. Since $\Delta(T \setminus e_2)$ is not a subset of a lawn, it is not defined and so not both $v_1, v_3 \in \bar{\Omega}_0$; thus we may assume (since we now have symmetry between e_1 and e_3) that v_1 is the middle vertex of some prominence P_1. Since e_2 is internal, it is drawn in $\Delta(P_1)$; but e_1 is external, and so $v_3 \in V(P_1)$, a contradiction as before. This completes the proof. □

The clockwise orientation of the wall provides a cyclic permutation Ω^* of the vertices of G drawn in it. Let G^* be the subgraph of G consisting of all vertices and edges drawn in the closed disc bounded by the wall. Then $(G^*, \Omega^*, \Omega_0)$ is a rural neighborhood, called the *park* of (G, Ω, Ω_0) (under the presentation $(\Sigma, \Gamma, \Delta, \Delta_0)$). Any rural neighborhood which can be constructed this way is called a *park*.

(11.4) *If $(G_0 \cup G^*, \Omega^*)$ is the composition of a park $(G^*, \Omega^*, \Omega_0)$ with a society (G_0, Ω_0), and (G_0, Ω_0) has no transaction of cardinality $> p$, then $(G_0 \cup G^*, \Omega^*)$ has no transaction of cardinality $> p + 2$.*

Proof. Choose $G, \Omega, \Sigma, \Gamma, \Delta, \Delta_0$ so that $(G^*, \Omega^*, \Omega_0)$ is the park of (G, Ω, Ω_0) under the presentation $(\Sigma, \Gamma, \Delta, \Delta_0)$. For any fence (A^*, B^*) of Ω^*, there is a fence (A, B) of Ω_0 and a set $Z \subseteq \bar{\Omega}_0$ such that

(i) $|Z| \leq 2$

(ii) $A^* \cap \bar{\Omega}_0 \subseteq A$, $B^* \cap \bar{\Omega}_0 \subseteq B$

(iii) for every lawn L, the set of all vertices of G^* drawn in $bd(L)$ is a subset of one of $A \cup A^* \cup Z$, $B \cup B^* \cup Z$. [p. 69] (This cannot necessarily be chosen with $Z = \emptyset$ because of the possibility that there are wall edges with one end in A^* and the other in B^*; but there are at most two such edges, and it follows easily that we may choose Z with $|Z| \leq 2$, as claimed.)

We claim that for every path P of $G_0 \cup G^*$ with one end in $A \cup A^*$ and the other in $B \cup B^*$ and no vertex in Z, there is a subpath P' of P with $P' \subseteq G_0$, with one end in A and the other in B. For there is certainly a subpath P' of P with one end in $A \cup A^*$, the other in $B \cup B^*$ and with no internal vertex in $A \cup A^* \cup B \cup B^*$. But then no internal vertex of P' is in $\bar{\Omega}_0$, and so either $P' \subseteq G^*$ or $P' \subseteq G_0$. If $P' \subseteq G^*$, then P' is drawn completely in some lawn L, since no internal vertex of P' is in $A \cup B$; but then the ends of P' are either both in $A \cup A^* \cup Z$ or both in $B \cup B^* \cup Z$, a contradiction. Thus $P' \subseteq G_0$, and so the ends of P' are in A and B, respectively, since $(A \cup A^*) \cap V(G_0) \subseteq A$ and $(B \cup B^*) \cap V(G_0) \subseteq B$. This proves our claim.

Now let $\mathcal{P}$ be a transaction in $(G_0 \cup G^*, \Omega^*)$ with fence (A^*, B^*). Then there exists $\mathcal{P}_0 \subseteq \mathcal{P}$ with $|\mathcal{P}_0| \geq |\mathcal{P}| - 2$ such that no member of $\mathcal{P}_0$ has a vertex in Z, since $|Z| \leq 2$. But for each $P \in \mathcal{P}_0$ there is a subpath P' of P in G_0 from A to B, as we saw above. Let $\mathcal{P}'$ be the set of all these P'; then $\mathcal{P}'$ is a transaction in (G_0, Ω_0) with fence (A, B), and $|\mathcal{P}'| \geq |\mathcal{P}| - 2$. The result follows. □

(11.5) *Let G be simple, and let (G, Ω) have sophistication $\leq p$. Then there is a society (G_1, Ω_1) and a rural neighborhood (G_2, Ω, Ω_1) such that*

(i) *$G_1, G_2 \subseteq G$, $G_1 \cup G_2 = G$ and $V(G_1 \cap G_2) = \bar{\Omega}_1$,*

(ii) *(G_1, Ω_1) has no transaction of cardinality $> p + 2$*

(iii) *for every triangle T of G, either $T \subseteq G_1$ or $T \subseteq G_2$.*

Proof. Choose a society (H_1, Ω_0) and a rural neighborhood (H_2, Ω, Ω_0) such that (G, Ω) is the composition of (H_1, Ω_0) with (H_2, Ω, Ω_0) and (H_1, Ω_0) has no transaction of cardinality $> p$. Let $(G^*, \Omega^*, \Omega_0)$ be the park of (H_2, Ω, Ω_0) under some presentation $(\Sigma, \Gamma, \Delta, \Delta_0)$. Let (G_1, Ω_1) be the composition of (H_1, Ω_0) with $(G^*, \Omega^*, \Omega_0)$; thus $\Omega_1 = \Omega^*$. Let G_2 be the sub-

graph of H_2 consisting of those vertices drawn in Σ either in the wall or in the complement in Δ of the disc bounded by the wall, and the wall edges and external edges of H_2. We claim that these satisfy the theorem. For (G_2, Ω, Ω_1) is a rural neighborhood, and

$$G_1 \cup G_2 = H_1 \cup G^* \cup G_2 = H_1 \cup H_2 = G;$$

$$V(G_1 \cap G_2) = V((H_1 \cup G^*) \cap G_2) = V(G^* \cap G_2) = \bar{\Omega}_1.$$

[p. 70] Thus (i) holds. (ii) holds because of (11.4). To verify (iii), let $T \subseteq G$ be a triangle. If $T \subseteq H_2$ then by (11.3) either no edge of T is internal, or no edge of T is external; that is, either $T \subseteq G_2$, or $T \subseteq G^* \subseteq G_1$, as required. Thus we may assume that $T \not\subseteq H_2$, and so some edge e of T is in $E(H_1)$. Thus every edge of $T \cap H_2$ has an end in $V(H_1)$, and so is not external, by (11.3). Hence $T \subseteq G_1$, as required. □

Let (G, Ω) be a society. A set $\mathcal{D}$ of separations of (G, Ω) is a *consolidation* of (G, Ω) if

(i) $|V(A \cap B)| \leq 3$ for all $(A, B) \in \mathcal{D}$

(ii) $A \subseteq B'$ for all distinct $(A, B), (A', B') \in \mathcal{D}$

(iii) $\bigcup(A : (A, B) \in \mathcal{D}) = G$

(iv) for each $(A, B) \in \mathcal{D}$ and every choice of distinct $u, v \in V(A \cap B)$ there is a path of A with ends u, v and with no internal vertex in B.

(11.6) *Every society has a consolidation.*

Proof. Let (G, Ω) be a society. For each vertex or edge x of G, let A_x be the subgraph of G consisting just of x (and its ends, if x is an edge). Let $B_x = G$ if x is a vertex, and let $B_x = G \setminus x$ if x is an edge. Then the set of all such (A_x, B_x) is a consolidation. □

(11.7) *If $\mathcal{D}$ is a consolidation of (G, Ω) and $C = \bigcap(B : (A, B) \in \mathcal{D})$ then $E(C) = \emptyset$, $\bar{\Omega} \subseteq V(C)$, $A \cap B \subseteq C$ for every $(A, B) \in \mathcal{D}$ and $A_1 \cap A_2 \subseteq C$ for every two distinct separations $(A_1, B_1), (A_2, B_2) \in \mathcal{D}$.*

Proof. For each edge $e \in E(G)$ there exists $(A, B) \in \mathcal{D}$ with $e \in E(A)$, and so $e \notin E(B)$; hence, $e \notin E(C)$ and so $E(C) = \emptyset$. For each $(A, B) \in \mathcal{D}$, $\bar{\Omega} \subseteq V(B)$ and so $\bar{\Omega} \subseteq V(C)$. Suppose that $(A, B) \in \mathcal{D}$. Then $A \cap B \subseteq B'$ for every $(A', B') \in \mathcal{D}$; for this is trivial if $(A, B) = (A', B')$, and otherwise $A \cap B \subseteq A \subseteq B'$. Thus $A \cap B \subseteq C$ follows. Finally, assume that $(A_1, B_1), (A_2, B_2) \in \mathcal{D}$ are distinct. Then $A_1 \cap A_2 \subseteq A_1 \cap B_1 \subseteq C$, by (ii) and the preceding part of (11.7), as required. □

Let $\mathcal{D}$ be a consolidation of (G, Ω). Its *core* is the simple graph C with $V(C) = \bigcap(V(B) : (A, B) \in \mathcal{D})$ in which distinct vertices u, v are adjacent if there exists $(A, B) \in \mathcal{D}$ with $u, v \in V(A \cap B)$. Note that $\bar{\Omega} \subseteq V(C)$.

(11.8) *Let $\mathcal{D}$ be a consolidation of (G,Ω) with core C. Let $C_1, C_2 \subseteq C$ with $C_1 \cup C_2 = C$ and $\bar{\Omega} \subseteq V(C_2)$. Let $\mathcal{D} = \mathcal{D}_1 \cup \mathcal{D}_2$, where $\mathcal{D}_1 \cap \mathcal{D}_2 = \emptyset$ and $V(A \cap B) \subseteq V(C_i)$ for $i = 1, 2$ and all $(A,B) \in \mathcal{D}_i$. Let C^* be the graph with $V(C^*) = V(C_1 \cap C_2)$ and $E(C^*) = \emptyset$. Let*

$$G_i = C^* \cup \bigcup (A : (A,B) \in \mathcal{D}_i) \qquad (i = 1, 2).$$

[p. 71] *Then (G_1, G_2) is a separation of (G,Ω), $G_1 \cap G_2 = C^*$, $V(G_1) \cap V(C) = V(C_1)$, and $V(G_2) \cap V(C) = V(C_2)$.*

Proof. (1) $G_1 \cup G_2 = G$.

For $G_1 \cup G_2 \subseteq G$, and

$$G = \bigcup (A : (A,B) \in \mathcal{D}) = \bigcup (A : (A,B) \in \mathcal{D}_1) \cup \bigcup (A : (A,B) \in \mathcal{D}_2) \subseteq G_1 \cup G_2.$$

(2) $G_1 \cap G_2 = C^*$.

Certainly $C^* \subseteq G_1 \cap G_2$. For the converse inclusion, it suffices to show that $A_1 \cap A_2 \subseteq C^*$ for $(A_1, B_1) \in \mathcal{D}_1$ and $(A_2, B_2) \in \mathcal{D}_2$. But $A_2 \subseteq B_1$, and so $A_1 \cap A_2 \subseteq A_1 \cap B_1 \subseteq C_1$, and similarly $A_1 \cap A_2 \subseteq C_2$; and hence $V(A_1 \cap A_2) \subseteq V(C_1 \cap C_2) = V(C^*)$; and $E(A_1 \cap A_2) = \emptyset$, and so $A_1 \cap A_2 \subseteq C^*$, as required.

(3) *$V(G_1) \cap V(C) = V(C_1)$ and $V(G_2) \cap V(C) = V(C_2)$.*

Let $X_i = V(G_i) \cap V(C)$ for $i = 1, 2$. Then

$$\begin{aligned} X_i &= V(C^*) \cup \bigcup (V(A \cap C) : (A,B) \in \mathcal{D}_i) \\ &\subseteq V(C_1) \cup \bigcup (V(A \cap B) : (A,B) \in \mathcal{D}_i) \end{aligned}$$

since $V(C^*) \subseteq V(C_i)$ and $V(C) \subseteq V(B)$ for each $(A,B) \in \mathcal{D}$. But $V(A \cap B) \subseteq V(C_i)$ for each $(A,B) \in \mathcal{D}_i$, and so $X_i \subseteq V(C_i)$. Using (1) and (2)

$$\begin{aligned} X_1 \cup X_2 &= V(G_1 \cup G_2) \cap V(C) = V(C) = V(C_1 \cup C_2), \\ X_1 \cap X_2 &= V(G_1 \cap G_2) \cap V(C) = V(C^*) = V(C_1 \cap C_2), \end{aligned}$$

which implies that $X_i = V(C_i)$ for $i = 1, 2$, as required for (3).

Now, $\bar{\Omega} \subseteq V(C_2) \subseteq V(G_2)$, by (3), and $E(C^*) = \emptyset$. Thus (1) and (2) imply (G_1, G_2) is a separation of (G,Ω). Along with (2) and (3) this completes the proof. □

(11.9) *Let $\mathcal{D}$ be a consolidation of (G,Ω) chosen so that its core C has as few vertices as possible. Then (C,Ω) is 4-connected.*

Proof. Suppose that (C,Ω) is not 4-connected. Then there is a 3-separation (C_1, C_2) of (C,Ω), and by choosing (C_1, C_2) with C_1 minimal,

we find that for every choice of distinct $u, v \in V(C_1 \cap C_2)$ there is a path of C_1 between u and v with no internal vertex in C_2; and no two vertices of $C_1 \cap C_2$ are adjacent in C_1. Let $\mathcal{D}_1 = \{(A, B) \in \mathcal{D} : V(A \cap B) \subseteq V(C_1)\}$, $\mathcal{D}_2 = \mathcal{D} - \mathcal{D}_1$. We claim that $V(A \cap B) \subseteq V(C_2)$ for all $(A, B) \in \mathcal{D}_2$. For since $(A, B) \notin \mathcal{D}_1$, there exists $v \in V(A \cap B) - V(C_1)$, and so by (11.7) with [p. 72] $v \in V(C_2)$. Since v is adjacent in C to every other member of $V(A \cap B)$, it follows that $V(A \cap B) \subseteq V(C_2)$, since $C_1 \cup C_2 = C$, as claimed. Thus we may define G_1, G_2 as in (11.8). We observe

(1) *For every choice of distinct $u, v \in V(G_1 \cap G_2)$ there is a path of G_1 between u and v with no internal vertex in G_2.*

Suppose for some choice of u, v there is no such path. Then there is a partition X, Y of $(V(G_1) - V(G_2)) \cup \{u, v\}$ with $u \in X, v \in Y$ such that no vertex in X is adjacent in G_1 to any vertex in Y. Now by choice of (C_1, C_2) there is a path of C_1 between u and v with no internal vertex in C_2, and by (11.8)

$$V(C_1) - V(C_2) \subseteq V(G_1) - V(G_2);$$

and so there exists $x \in X$ and $y \in Y$, adjacent in C_1. Choose $(A, B) \in \mathcal{D}$ with $x, y \in V(A \cap B)$. Now not both $x, y \in V(C_1 \cap C_2)$, since by our choice of (C_1, C_2) no two vertices of $C_1 \cap C_2$ are adjacent in C_1. Thus $V(A \cap B) \not\subseteq V(C_2)$, and so $(A, B) \in \mathcal{D}_1$, and $A \subseteq G_1$. But there is a path P of A between x and y with no internal vertex in B. Since $V(P) \subseteq V(A) \subseteq V(G_1)$ and since, by (11.8),

$$V(G_1 \cap G_2) = V(C_1 \cap C_2) \subseteq V(C) \subseteq V(B)$$

by the definition of "core," it follows that no internal vertex of P belongs to $V(G_2)$. Therefore $V(P) \subseteq X \cup Y$, and so there is a vertex of P in X, and a vertex of P in Y which are adjacent in P and hence in G_1. This is a contradiction, as required.

(2) *For every $(A_2, B_2) \in \mathcal{D}_2$, $A_2 \subseteq G_2$ and $G_1 \subseteq B_2$.*

For the first inclusion holds by definition of G_2. The second holds since $V(C_1 \cap C_2) \subseteq V(B_2)$ and $A \subseteq B_2$ for every $(A, B) \subseteq \mathcal{D}_1$, since $\mathcal{D}$ is a consolidation.

Let $\mathcal{D}' = \mathcal{D}_2 \cup \{(G_1, G_2)\}$. We deduce from (1) and (2) and (11.8) that

(3) *$\mathcal{D}'$ is a consolidation of (G, Ω).*

Let its core be C'.

(4) $V(C') \subseteq V(C)$.

For let $v \in V(C')$. Then $v \in V(B)$ for every $(A, B) \in \mathcal{D}_2$, and $v \in V(G_2)$. If $v \in V(C_1 \cap C_2)$ then certainly $v \in V(C)$. If not, then there exists

$(A, B) \in \mathcal{D}_2$ with $v \in V(A)$, by definition of G_2, and so $v \in V(A \cap B)$. But $V(A \cap B) \subseteq V(C)$, by (11.7), and so $v \in V(C)$, as required.

(5) *C' has fewer vertices than C.*

[p. 73] For since (C_1, C_2) is a 3-separation of (C, Ω), there exists $v \in V(C_1) - V(C_2)$. Since $C_1 \subseteq C$ and $V(C') \subseteq V(G_2)$ it follows that

$$V(C_1) \cap V(C') \subseteq V(C) \cap V(G_2) = V(C_2)$$

by (11.8) and therefore $v \notin V(C')$. Hence $V(C) \neq V(C')$, and the result follows from (4).

But (5) contradicts our initial choice of $\mathcal{D}$, and completes the proof. □

The pattern of a transaction $\mathcal{P}$ in (G, Ω) is the set $\{X(P) : P \in \mathcal{P}\}$ where $X(P)$ denotes the set of ends of $P \in \mathcal{P}$.

(11.10) *Let C be the core of a consolidation $\mathcal{D}$ of (G, Ω). For any Π, there is a transaction in (G, Ω) with pattern Π if and only if there is such a transaction in (C, Ω).*

Proof. A jump is a path in G with distinct ends, both in $V(C)$, and with no internal vertex in $V(C)$. If Q is a jump then by (iii) and the last part of (11.7) there exists $(A, B) \in \mathcal{D}$ with $Q \subseteq A$, and hence its ends are adjacent in C; we define $f(Q)$ to be the 1-edge path of C joining these ends. Every Ω-path P in (G, Ω) decomposes uniquely into a concatenation $Q_1 \cup \cdots \cup Q_k$ of subpaths each of which are jumps; we define $f(P)$ to be the concatenation of $f(Q_1), \ldots, f(Q_k)$. Then $f(P)$ is an Ω-path in C. If $\mathcal{P} = \{P_1, \ldots, P_p\}$ is a transaction in (G, Ω) we define $f(\mathcal{P})$ to be $\{f(P_1), \ldots, f(P_p)\}$; then $f(\mathcal{P})$ is a transaction in (C, Ω) with the same pattern as $\mathcal{P}$. This proves the "only if" part of the lemma.

For the "if" part we proceed as follows. For each edge e of C, there exists $(A, B) \in \mathcal{D}$ with both ends of e in $V(A \cap B)$; let $g(e)$ be a jump in A between the ends of e. (This exists since $\mathcal{D}$ is a consolidation.) Note that $g(e_1)$ meets $g(e_2)$ only if $e_1, e_2 \in E(C)$ share an end. For, there are (A_1, B_1), $(A_2, B_2) \in \mathcal{D}$ such that e_i joins distinct vertices of $A_i \cap B_i$ and $g(e_i)$ is a jump in A_i between these vertices, for $i = 1, 2$. If $(A_1, B_1) = (A_2, B_2)$ then e_1, e_2 share an end by (i). If not then $V(g(e_1) \cap g(e_2)) \subseteq V(A_1 \cap A_2) \subseteq V(C)$ by (11.7) and so, since $g(e_1)$ and $g(e_2)$ are jumps, any common vertices of $g(e_1)$ and $g(e_2)$ must be common ends of $g(e_1)$ and $g(e_2)$, and therefore also of e_1, e_2.

For each Ω-path P of (C, Ω), the graph $\bigcup(g(e) : e \in E(P))$ is connected, and so includes a path of G between the ends of P. We define $g(P)$ to be such a path. For disjoint Ω-paths P_1, P_2 of (C, Ω), the paths $g(P_1), g(P_2)$ of G are also disjoint, for $g(e_1)$ meets $g(e_2)$ only if $e_1, e_2 \in E(C)$ share an end.

Thus if $\mathcal{P} = \{P_1, \ldots, P_p\}$ is a transaction in (C, Ω) then $\{g(P_1), \ldots, g(P_p)\}$ is a transaction in (G, Ω) with the same pattern. This completes the proof. □

[p. 74] With these lemmas in hand, we turn to the main theorem of this section. A segregation of a society (G, Ω) is a set $\{(A_0, \Omega_0), \ldots, (A_k, \Omega_k)\}$ of societies such that

(i) $A_0 \cup \cdots \cup A_k = G$ and $E(A_i \cap A_j) = \emptyset$ for $0 \le i, j \le k$ with $i \ne j$

(ii) $V(A_i \cap A_j) \subseteq \bar{\Omega}_i$ for $0 \le i, j \le k$ with $i \ne j$

(iii) $\bar{\Omega} \subseteq \bar{\Omega}_0 \cup \cdots \cup \bar{\Omega}_k$.

An arrangement of a segregation $\{(A_0, \Omega_0), \ldots, (A_k, \Omega_k)\}$ of (G, Ω) in a closed disc Δ is a function α with domain

$$\{(A_0, \Omega_0), \ldots, (A_k, \Omega_k)\} \cup \bar{\Omega}_0 \cup \cdots \cup \bar{\Omega}_k$$

and with the following properties:

(i) for $0 \le i \le k$, $\alpha((A_i, \Omega_i))$ is a closed disc $\Delta_i \subseteq \Delta$ and $\alpha(v) \in bd(\Delta_i)$ for each $v \in \bar{\Omega}_i$, and $\alpha(v) \in bd(\Delta)$ for each $v \in \bar{\Omega}$

(ii) for $0 \le i < j \le k$, if $x \in \Delta_i \cap \Delta_j$, then $x = \alpha(v)$ for some $v \in \bar{\Omega}_i \cap \bar{\Omega}_j$, and for $0 \le i \le k$, if $x \in \Delta_i \cap bd(\Delta)$ then $x = \alpha(v)$ for some $v \in \bar{\Omega}_i \cap \bar{\Omega}$

(iii) $\alpha(u) \ne \alpha(v)$ for distinct $u, v \in \bar{\Omega}_0 \cup \cdots \cup \bar{\Omega}_k$

(iv) for $0 \le i \le k$, Ω_i is the cyclic permutation of $\bar{\Omega}_i$ derived (under α^{-1}) from the clockwise orientation of $bd(\Delta_i)$, and Ω is the permutation derived from the clockwise orientation of $bd(\Delta)$.

A segregation is *feasible* if there is an arrangement of it in some disc. The main result of this section is the following:

(11.11) *Let $p \ge 2$, with $p \ne 3$. Let (G, Ω) be a society with no crooked transaction of cardinality p. Then there is a feasible segregation $\{(A_0, \Omega_0), \ldots, (A_k, \Omega_k)\}$ of (G, Ω) such that $|\bar{\Omega}_1|, \ldots, |\bar{\Omega}_k| \le 3$ and (A_0, Ω_0) has no transaction of cardinality $> 3p + 9$.*

Proof. By (11.6) there is a consolidation of (G, Ω). Let us choose a consolidation $\mathcal{D}$ such that its core has as few vertices as possible, and subject to that $|\mathcal{D}|$ is minimum. It follows that

(1) *If $(A, B), (A', B') \in \mathcal{D}$ are distinct then $V(A \cap B) \not\subseteq V(A')$.*

For suppose that $V(A \cap B) \subseteq V(A')$. Let

$$\mathcal{D}' = (\mathcal{D} - \{(A, B), (A', B')\}) \cup \{(A \cup A', B \cap B')\}.$$

We claim that $\mathcal{D}'$ is a consolidation of (G, Ω). For $(A \cup A') \cap (B \cap B') = A' \cap B'$ since $A' \subseteq B$ and $A \cap B \subseteq A'$, and so conditions (i), (iii), (iv) in

the definition of a consolidation are easily verified. For condition (ii), let $(A'', B'') \in \mathcal{D} - \{(A, B), (A', B')\}$; we must show that $A'' \subseteq B \cap B'$ and [p. 75] $A \cup A' \subseteq B''$. But $A'' \subseteq B$, $A'' \subseteq B'$, $A \subseteq B''$ and $A' \subseteq B''$ since $\mathcal{D}$ is a consolidation, and so both inclusions follow. Thus $\mathcal{D}'$ is a consolidation of (G, Ω). Since $V(A \cap B) \subseteq V(A' \cap B')$ (because $A \subseteq B'$ and $A \cap B \subseteq A'$) we deduce that the cores of $\mathcal{D}$ and $\mathcal{D}'$ have the same vertex sets. Yet $|\mathcal{D}'| < |\mathcal{D}|$ contrary to our choice of $\mathcal{D}$. This proves (1).

Let C be the core of $\mathcal{D}$. By (11.10), (C, Ω) has no crooked transaction of cardinality p, and by (11.9) (C, Ω) is 4-connected. By (6.1) (C, Ω) has sophistication at most $3p + 7$. By (11.5) there is a society (C_1, Ω_0) and a rural neighborhood (C_2, Ω, Ω_0) such that

(i) $C_1, C_2 \subseteq C$, $C_1 \cup C_2 = C$ and $V(C_1 \cap C_2) = \bar{\Omega}_0$

(ii) (C_1, Ω_0) has no transaction of cardinality $> 3p + 9$

(iii) for every triangle T of C, either $T \subseteq C_1$ or $T \subseteq C_2$.

Choose $\mathcal{D}_1, \mathcal{D}_2 \subseteq \mathcal{D}$ with $\mathcal{D}_1 \cup \mathcal{D}_2 = \mathcal{D}$ and $\mathcal{D}_1 \cap \mathcal{D}_2 = \emptyset$, so that for $i = 1, 2$ and all $(A, B) \in \mathcal{D}_i$, $V(A \cap B) \subseteq V(C_i)$ and every two distinct members of $V(A \cap B)$ are adjacent in C_i. Let C^* be the graph with $V(C^*) = V(C_1 \cap C_2)$ and $E(C^*) = \emptyset$. Let

$$A_0 = C^* \cup \bigcup (A : (A, B) \in \mathcal{D}_1)$$

$$B_0 = C^* \cup \bigcup (A : (A, B) \in \mathcal{D}_2).$$

From (11.8) we see that

(2) *(A_0, B_0) is a separation of (G, Ω), and $V(A_0 \cap B_0) = \bar{\Omega}_0$, and $V(B_0 \cap C) = V(C_2)$.*

For each $v \in V(C^*)$ let X_v be the subgraph of A_0 with only one vertex v and no edges. Let

$$\mathcal{D}'_1 = \{(A, A_0 \cap B) : (A, B) \in \mathcal{D}_1\} \cup \{(X_v, A_0) : v \in V(C^*)\}.$$

(3) *$\mathcal{D}'_1$ is a consolidation of (A_0, Ω_0).*

For if $(A, B) \in \mathcal{D}_1$, then $A \subseteq A_0$ by definition of A_0, and

$$\bar{\Omega}_0 = V(C^*) \subseteq V(A_0) \cap V(C) \subseteq V(A_0 \cap B);$$

and so $\mathcal{D}'_1$ is a set of separations of (A_0, Ω_0). We verify conditions (i), ..., (iv) in the definition of a "consolidation." Condition (iii) holds from the definition of A_0. Conditions (i) and (iv) hold since $A \cap B = A \cap (A_0 \cap B)$ for $(A, B) \in \mathcal{D}_1$ (because $A \subseteq A_0$). For condition (ii), let $(A', B'), (A'_1, B'_1) \in \mathcal{D}'_1$ be distinct. We shall show that $A' \subseteq B'_1$. If $(A'_1, B'_1) = (X_v, A_0)$ for some

$v \in V(C^*)$ then $A' \subseteq B'_1$ since $A' \subseteq A_0$; and so [p. 76] we may assume that $(A'_1, B'_1) = (A_1, A_0 \cap B_1)$ for some $(A_1, B_1) \in \mathcal{D}_1$. If $(A', B') = (X_v, A_0)$ for some $v \in V(C^*)$ then $A' \subseteq B'_1$ since $v \in \bar{\Omega}_0$ and (A'_1, B'_1) is a separation of (A_0, Ω_0); and so we may assume that $(A', B') = (A, A_0 \cap B)$ for some $(A, B) \in \mathcal{D}_1$. But then $(A, B) \neq (A_1, B_1)$, and so $A \subseteq B_1$ since $\mathcal{D}$ is a consolidation, and $A \subseteq A_0$, and so $A \subseteq A_0 \cap B_1$; thus $A' \subseteq B'_1$ as required.

(4) (A_0, Ω_0) *has no transaction of cardinality* $> 3p + 9$.

For let C' be the core of $\mathcal{D}'_1$. For any $v \in V(C')$, $v \in V(A_0)$ and $v \in V(B)$ for every $(A, B) \in \mathcal{D}_1$. Since $v \in V(A_0)$, either $v \in V(C^*)$ or $v \in V(A)$ for some $(A, B) \in \mathcal{D}_1$. We deduce that either $v \in V(C^*)$ or $v \in V(A \cap B) \subseteq V(C_1)$ for some $(A, B) \in \mathcal{D}_1$. In either case $v \in V(C_1)$, and so $V(C') \subseteq V(C_1)$. We claim that $E(C') \subseteq E(C_1)$. For let $u, v \in V(C')$ be adjacent in C'. Then there exists $(A', B') \in \mathcal{D}'_1$ with $u, v \in V(A' \cap B')$. Hence $|V(A')| \geq 2$, and so $(A', B') = (A, A_0 \cap B)$ for some $(A, B) \in \mathcal{D}_1$, and it follows from the definition of $\mathcal{D}_1$ that u, v are adjacent in C_1. Thus $E(C') \subseteq E(C_1)$ as claimed, and so $C' \subseteq C_1$. Since (C_1, Ω_0) has no transaction of cardinality $> 3p + 9$, nor does (C', Ω_0), and hence by (11.10) (A_0, Ω_0) has no transaction of cardinality $> 3p + 9$.

Now (C_2, Ω, Ω_0) is a rural neighborhood. Take a presentation $(\Sigma, \Gamma, \Delta, \Delta_0)$ of it, and for each $v \in V(C_2)$ let $\alpha(v)$ be the image under Γ of v. Since (C, Ω) is simple and 4-connected, the image under Γ of any triangle of C_2 bounds a region of the drawing. By a ≤ 3-clique we mean a complete subgraph of C_2 with ≤ 3 vertices. It follows that we may choose a closed disc $\beta(K) \subseteq \Delta$ for each ≤ 3-clique K such that

(i) for every ≤ 3-clique K, if $x \in \beta(K) \cap \Delta_0$ then $x = \alpha(v)$ for some $v \in V(K \cap A_0)$, and if $x \in \beta(K) \cap bd(\Delta)$ then $x = \alpha(v)$ for some $v \in V(K) \cap \bar{\Omega}$

(ii) for distinct ≤ 3-cliques K, K', if $x \in \beta(K) \cap \beta(K')$ then $x = \alpha(v)$ for some $v \in V(K \cap K')$

(iii) for every ≤ 3-clique K and $v \in V(C_2)$, $\alpha(v)$ is not in the interior of $\beta(K)$

(iv) for every ≤ 3-clique K and $v \in V(C_2)$, $\alpha(v) \in bd(\beta(K))$ if and only if $v \in V(K)$.

Let $\mathcal{D}_2 = \{(A_1, B_1), \ldots, (A_k, B_k)\}$. For $1 \leq i \leq k$, let K_i be the clique of C_2 with $V(K_i) = V(A_i \cap B_i)$. By (1), $K_1, \ldots, K_k$ are all distinct. For $1 \leq i \leq k$, let Ω_i be the cyclic permutation of $V(K_i)$ induced by the clockwise orientation of $bd(\beta(K_i))$.

(5) $\{(A_0, \Omega_0), (A_1, \Omega_1), \ldots, (A_k, \Omega_k)\}$ *is a segregation of* (G, Ω).

To show this we verify conditions (i), (ii), (iii) in the definition of a "segregation."

[p. 77] *Condition (i).* First,

$$A_0 \cup A_1 \cup \cdots \cup A_k = C^* \cup \bigcup(A : (A,B) \in \mathcal{D}_1) \cup \bigcup(A : (A,B) \in \mathcal{D}_2)$$
$$\supseteq \bigcup(A : (A,B) \in \mathcal{D}) = G.$$

Second, for $(A,B) \in \mathcal{D}_1$ and $1 \le j \le k$, $E(A \cap A_j) = \emptyset$ since (A,B), (A_j,B_j) are distinct members of the consolidation $\mathcal{D}$, and so $E(A_0 \cap A_j) = \emptyset$. Third, for $1 \le i,j \le k$ with $i \ne j$, $E(A_i \cap A_j) = \emptyset$ since (A_i,B_i), (A_j,B_j) are distinct members of $\mathcal{D}$.

Condition (ii). First, for $(A,B) \in \mathcal{D}_1$ and $1 \le j \le k$, $A \subseteq B_j$ and $A_j \subseteq B$ and so

$$V(A \cap A_j) = V(A \cap B) \cap V(A_j \cap B_j) \subseteq V(C_1) \cap V(C_2) \cap \bar{\Omega}_j = \bar{\Omega}_0 \cap \bar{\Omega}_j.$$

Moreover, $V(C^*) \subseteq V(B_j)$ and so $V(C^*) \cap V(A_j) \subseteq \bar{\Omega}_0 \cap V(A_j \cap B_j) = \bar{\Omega}_0 \cap \bar{\Omega}_j$. Hence for $1 \le j \le k$, $V(A_0 \cap A_j) \subseteq \bar{\Omega}_0 \cap \bar{\Omega}_j$. Second, for $1 \le i,j \le k$ with $i \ne j$, $V(A_i \cap A_j) \subseteq V(A_i \cap B_i) = \bar{\Omega}_i$ since $A_j \subseteq B_i$.

Condition (iii). Let $v \in \bar{\Omega}$. We shall show that $v \in \bar{\Omega}_i$ for some i with $0 \le i \le k$. Since $\mathcal{D}$ is consolidation, there exists $(A,B) \in \mathcal{D}$ with $v \in V(A)$. Hence either $(A,B) \in \mathcal{D}_2$ or $v \in V(A_0)$, and so there exists i with $0 \le i \le k$ with $v \in V(A_i)$. But $\bar{\Omega} \subseteq V(B_0)$ by (2), and $\bar{\Omega} \subseteq V(B_j)$ for $1 \le j \le k$ since each $(A_j,B_j) \in \mathcal{D}$ is a separation of (G,Ω). Hence $v \in V(B_i)$ and so $v \in V(A_i \cap B_i) = \bar{\Omega}_i$. It follows that $\bar{\Omega} \subseteq \bar{\Omega}_0 \cup \bar{\Omega}_1 \cup \cdots \cup \bar{\Omega}_k$ as required.

This completes the proof of (5).

Let us define $\alpha(v)$ to be the drawing of v for each $v \in \bar{\Omega}_0 \cup \cdots \cup \bar{\Omega}_k$, and define $\alpha((A_i,\Omega_i)) = \beta(K_i)$ for $1 \le i \le k$ and $\alpha((A_0,\Omega_0)) = \Delta_0$. Then α is an arrangement of the segregation, and the theorem follows. □

References

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